

Large-Area Satellite-Derived Urban Reconstruction and Manifest-Driven Integration into ChordAtlas

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MSc Computer Graphics, Vision and Imaging

This report is submitted as part requirement for the MSc Degree in 'Computer Graphics, Vision & Imaging', at University College London. It is substantially the result of my own work except where explicitly indicated in the text.

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Abstract

Recent AI-based approaches, particularly deep-learning models, can reconstruct three-dimensional urban geometry from satellite imagery. But bounded inference windows, discontinuities between overlapping predictions, incomplete large-area appearance output, and limited geospatial interoperability constrain regional use. This dissertation develops and evaluates a hybrid workflow that extends Sat3DGen large-image inference and connects its outputs to open geospatial data and ChordAtlas.

The proposed large-image method processes a rectangular satellite mosaic with overlapping windows, maps their predictions into a shared density field at sub-voxel precision, and blends them with raised-cosine weights before a single surface extraction. A memory-bounded second pass reconstructs the window features and blends per-vertex colour on the fixed global mesh. In a London case study, this reduced mean density discontinuities at the former window boundaries by 49–60 per cent and mean top-surface discontinuities by 60–75 per cent relative to a matched hard-crop baseline. Every extracted vertex received colour support without changing the mesh geometry.

OpenStreetMap footprints and a one-metre Environment Agency digital surface model then provide geographic identity, semantic context, and a common vertical reference. The resulting assets are cropped by footprint, retain vertex RGB, and are published through explicit coordinate, manifest, cache, and validation contracts to the ChordAtlas environment. A separate mesh-space study evaluates the behaviour of independent-tile stitching and DSM correction. Together, the results establish a large-area reconstruction and asset-integration framework that combines learned density and appearance with explicit spatial and publication contracts for editable urban content.

Keywords— urban reconstruction - satellite imagery - large-image inference - density fusion - vertex colour - ChordAtlas - geospatial integration

To my family and everyone who supported this work.

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The computer was born to solve problems that
did not exist before.

— Bill Gates

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I am grateful to Professor Anthony Steed for his supervision and for shaping the project from a broad world-generation theme into a concrete investigation of geographically steered urban reconstruction.

Declaration

I, Renjie Li, declare that the thesis has been composed by myself and that the work has not been submitted for any other degree or professional qualification. I confirm that the work submitted is my own, except where work which has formed part of jointly-authored publications has been included and referenced. The report may be freely copied and distributed provided the source is explicitly acknowledged.

14/08/26

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Date

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List of Abbreviations

AOI: Area of Interest

API: Application Programming Interface

BNG: British National Grid

CRS: Coordinate Reference System

DSM: Digital Surface Model

GIS: Geographic Information System

GUI: Graphical User Interface

LiDAR: Light Detection and Ranging

OBJ: Wavefront Object Mesh Format

OSM: OpenStreetMap

WGS84: World Geodetic System 1984

1 | Introduction

1.1 Motivation and problem statement

Deep generative models conditioned on widely available satellite imagery offer a promising route to detailed urban geometry. Their outputs are not, however, immediately suitable for an editable city-modelling environment. Inference is performed over bounded image windows; neighbouring predictions may disagree in position, height, and topology; and a mesh file does not by itself identify its geographic extent, building ownership, coordinate frame, or processing history. These issues become particularly important when the output must support procedural editing or interactive use rather than a single rendered view. Figure 1.1 illustrates the form of one retained local output.

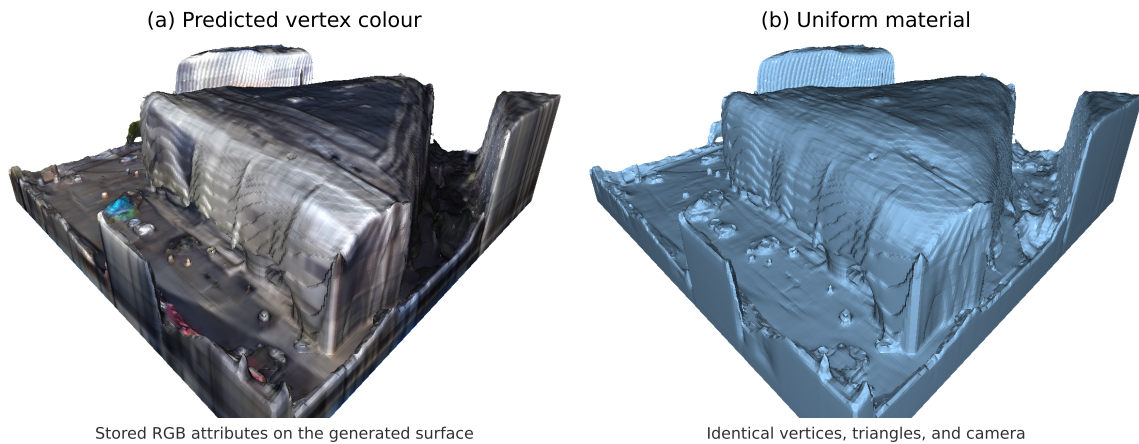


Figure 1.1: Representative Sat3DGen output for a London inference tile centred at 51.510303° N, 0.133978° W. Panel (a) renders the stored predicted vertex colours; panel (b) renders the same vertices and triangles with a uniform material. The paired views separate predicted appearance from the generated surface in the model’s normalised local frame.

Procedural city systems address a complementary part of the problem. Building footprints, semantic parts, and explicit rules provide stable identity and authoring control,

but do not recover the observed appearance and geometric detail of a real location automatically. This dissertation studies the interface between these representations. It asks how learned urban geometry can be combined with open geospatial context and delivered as building-level assets to ChordAtlas, an existing procedural urban-modelling environment [1].

A case study is centred on London and uses Sat3DGen as the satellite-conditioned generator [2]. The study extends its large-image workflow with continuous density placement, smooth overlap weighting, and vertex-colour recovery before connecting the resulting surface to downstream geospatial processing. OpenStreetMap (OSM) provides footprints and contextual semantics, while an Environment Agency digital surface model (DSM) supplies an external vertical reference [3, 4]. The resulting problem is both geometric and operational: tiles must be placed and combined coherently, and every published asset must retain enough spatial and processing metadata to be interpreted by the downstream system.

1.2 Research aim and question

The aim is to develop and evaluate a hybrid machine-learning/procedural workflow that reconstructs a contiguous region from overlapping satellite-conditioned predictions and converts the resulting geometry into coordinate-explicit, building-associated assets for ChordAtlas. The study addresses the following research question:

To what extent can overlapping satellite-conditioned predictions be combined into a coloured regional mesh and integrated with geospatial context in ChordAtlas through a manifest-driven, coordinate-explicit workflow?

The question treats integration as more than file conversion. A useful result must establish a consistent spatial frame, associate geometry with geographic features, preserve

appearance information, expose processing state, and support repeatable transfer between Python mesh processing and the Java-based ChordAtlas environment.

1.3 Design rationale and evaluated scope

The broader project theme concerned geographically steered world generation for interactive and collaborative environments. Sat3DGen was selected because it produces explicit coloured geometry from satellite imagery, making it a more direct input to an editing system than an image-only generator. Initial feasibility work established local model inference and explored Unity/Ubiq integration. A London cross-view collection was then prepared to investigate location-specific data requirements.

Resource analysis showed that reproducing the released Sat3DGen training regime would dominate the available computational budget. The empirical study therefore retains the released model and evaluates two complementary forms of regional assembly. The first adapts the released large-image wrapper so that rectangular mosaics are fused smoothly in density space and colour is recovered on the global surface. The second places independently extracted meshes in a shared geographic frame and evaluates stitching, height reconciliation, and building association. Both paths subsequently enter the same publication architecture.

OSM and DSM data serve different roles in this design. Footprints provide building identity and a basis for per-building extraction; the DSM provides a metric surface-height reference for vertical processing. ChordAtlas provides the procedural destination, including its tiled MiniMesh representation and building-generation workflow. BigSUR and FrankenGAN inform the wider goal of structured façade modelling, while CityEngine provides a procedural workflow comparator and Ubiq motivates future collaborative deployment [5, 6, 7, 8]. These systems establish the architectural context for the study; the evaluated contribution comprises the London data study, large-image density and colour

method, mesh-processing experiments, and ChordAtlas integration path.

1.4 Objectives

Six objectives make the research question operational:

1. construct and characterise a London multimodal case study comprising satellite imagery, panoramas, OSM context, and DSM data;
2. adapt and evaluate large-image density fusion and per-vertex colour recovery while retaining the released Sat3DGen model;
3. define the coordinate, tiling, and metadata contracts that connect geographic inputs, Sat3DGen density and mesh spaces, and ChordAtlas;
4. implement mesh-space placement, cropping, overlap processing, DSM-informed height correction, and footprint-based building extraction;
5. implement static and on-demand publication of building assets to ChordAtlas while preserving stable identity and vertex colour; and
6. evaluate data integrity, geometric behaviour, topology, coordinate consistency, and cross-language publication, and relate the findings to Sat3DGen and procedural city-generation methods.

1.5 Contributions

The first contribution is a London data-preparation and integrity study that connects cross-view imagery with OSM and DSM context. It quantifies geographic support, channel availability, duplicate imagery, split relationships, and candidate-label geometry. These measurements show which properties must be checked before a geographically assembled collection is used for learning or evaluation.

The second contribution is a rectangular large-image inference method. It places overlapping density predictions at sub-voxel coordinates, blends them with smooth spatial weights, extracts one global surface, and recovers per-vertex RGB in a memory-bounded second pass. A matched experiment measures the reduction in window-boundary discontinuity relative to hard-crop fusion.

The third contribution is a mesh-space integration method for placing and combining independently generated tiles. The method introduces explicit coordinate conversion, footprint association, DSM-informed vertical processing, and per-building extraction. Controlled geometry cases and topological measurements evaluate the assumptions made by bottom removal, stitching, and height correction.

The fourth contribution is a ChordAtlas bridge that publishes selected geometry as modular building assets. Stable request identities, manifests, validation gates, staged publication, vertex-colour transfer, and Python–Java conversion make the data boundary inspectable and repeatable. The evaluation identifies the conditions under which this bridge operates reliably and the geometry and transaction properties that most strongly affect regional extension.

Sat3DGen, ChordAtlas, BigSUR, FrankenGAN, CityEngine, and Ubiq are pre-existing systems. The project-specific work reported here comprises the London data workflow, large-image density and colour extension, geospatial mesh-processing method, ChordAtlas integration, and their evaluation. External software, datasets, and published methods are identified and cited where they are used.

1.6 Dissertation structure

Chapter 2 reviews procedural urban modelling, satellite-conditioned reconstruction, semantic city representations, geospatial contracts, and related systems. Chapter 3 defines the data, large-image inference, coordinate, mesh-processing, publication, and evaluation

methods. Chapter 4 reports the data-integrity, density-fusion, colour, geometry, coordinate, and ChordAtlas integration results. Chapter 5 interprets the findings, compares the approach with Sat3DGen and CityEngine, and evaluates validity and future directions. Chapter 6 summarises the research outcome and contributions. The appendix records implementation provenance and detailed component-test results.

2 | Background and Related Work

2.1 Procedural and data-driven urban modelling

Procedural modelling represents architecture through rules and parameters rather than solely through sampled surfaces. Early city systems used extended L-systems to generate street networks and building populations [9]; split grammars then showed how attributes and rules could control architectural variation at scale [10]. CGA shape grammar made the relationship between initial shapes, rule operations, and polygonal buildings explicit [11]. ProceX made procedural extrusions interactive: footprint-driven operations create architectural massing that can be edited rather than merely viewed [12]. This structured representation is attractive for planning, simulation, visualisation, and content authoring because it preserves semantic parts.

Open building-footprint collections provide initial shapes for applying this principle beyond manually authored scenes. Microsoft’s Global ML Building Footprints dataset supplies predicted building outlines at large geographic scale [13]. These outlines provide location and plan extent, while height, appearance, and architectural semantics must still be reconstructed or generated.

Real data rarely arrives as clean procedural parameters. BigSUR addresses a closely related integration problem by combining noisy large-scale meshes, imagery, and geographic information system (GIS) footprints to recover structured urban models [5]. FrankenGAN then enriches coarse mass models with semantic façade detail and appearance through style-synchronised generative networks [6]. ChordAtlas exposes these research components in an interactive Java environment, with generators for GIS data, panoramas, meshes, and blocks [1]. FrankenGAN assumes a structured building-mass representation: it is neither a topology-repair method for arbitrary meshes nor a source of ground-truth façade texture. ChordAtlas appearance exports are therefore treated here as procedurally

generated assets rather than texture-preserving conversions of Sat3DGen triangles.

Recent geospatial approaches divide geometry and appearance processing in different ways. GeoTexBuild begins with map footprints and separates height-map generation, geometry reconstruction, and appearance stylisation [14]. Buyukdemircioglu and Oude Elberink instead texture existing CityJSON geometry from oblique and nadir aerial imagery [15]. Their contrasting starting points clarify that geometric reconstruction, structured city representation, and appearance transfer are related but distinct stages.

ArcGIS CityEngine is a relevant industrial authoring system in this procedural lineage. Its central workflow combines streets, blocks, lots, initial shapes, GIS attributes, assets, and authored CGA rules to generate polygonal city content [7]. File-geodatabase attributes can be retained and used to drive rules, while the documented runtime and SDK support scripted generation and export [16, 17]. These capabilities favour controllable regeneration, scenario authoring, and attribute-aware content production. They do not, however, imply automatic recovery of a real building from satellite pixels. Likewise, importing a dense OBJ does not by itself create stable building identity or roof, wall, and window semantics. CityEngine therefore serves as a representation and workflow comparator, including an executed local example, but not as a satellite-reconstruction accuracy baseline.

This lineage establishes two requirements for the present integration. First, generated geometry must be spatially compatible with footprints and panoramas, not merely visually plausible in isolation. Secondly, it must be converted into ChordAtlas’s tiled MiniMeshes and generator graph without corrupting an existing project when conversion fails. The baseline ChordAtlas workflow can clip an already loaded mesh and construct a block. The proposed path adds external, on-demand generation and publication, so state and failure handling become part of the research problem.

2.2 Satellite-conditioned street-level geometry

Satellite-to-ground synthesis is underconstrained: roofs and plan-view context are visible from above, while façades and street-level occlusions are not. Cross-view datasets reduce that ambiguity by pairing overhead imagery with ground panoramas. VIGOR formalised a setting in which a query is not necessarily centred in its reference satellite image and uses overlapping reference tiles [18]. Its correspondence geometry is therefore not an incidental file format. Candidate offsets must describe where the ground query lies within each aerial tile, and sufficient overlap is required if several candidates are meant to contain the same location.

Sat2Density learns a neural density representation from satellite-ground pairs [19]. Its successor extends the cross-view rendering lineage [20]. Sat2Scene uses diffusion for urban scene generation [21], while Sat2City uses a cascaded latent diffusion formulation for city generation [22]. These methods are related baselines, but their tasks, training data, and output representations differ. The comparison therefore considers their architectures and evaluation protocols rather than treating their reported metrics as interchangeable.

At the broader systems level, Meta’s WorldGen and World Labs illustrate a move towards prompt- and observation-conditioned 3D environments [23, 24]. Their relevance here is the objective of producing navigable and editable world content from location-specific observations; they provide systems context rather than directly comparable Sat3DGen baselines.

Sat3DGen places particular emphasis on explicit geometry. Its published pipeline uses a frozen DINOv3 image encoder, spatial padding tokens, a decoder that produces three feature planes, and a small network that predicts density and colour [25, 2]. The tri-plane design is related to efficient geometry-aware neural rendering [26]. Training adds a gravity-based density-variation loss, a relative-depth prior derived from Depth Anything

V2 [27], and randomly projected perspective views from ground panoramas. The density representation can be rendered from satellite or street-level viewpoints and converted into an explicit surface through Marching Cubes [2, 28]. Figure 2.1 separates this published model from the large-image and geospatial extensions studied here.

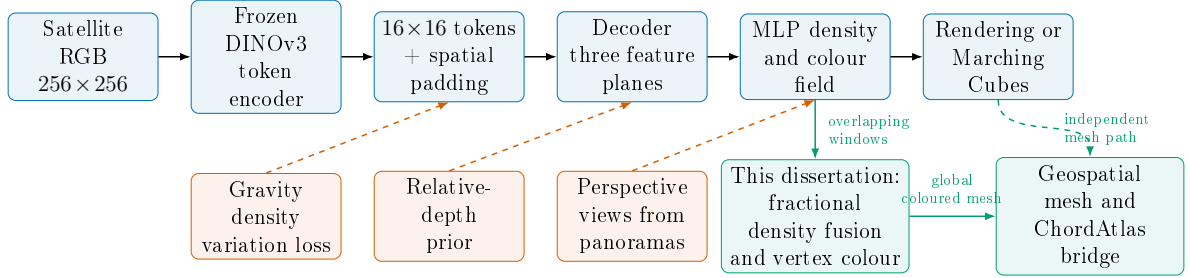


Figure 2.1: Published Sat3DGen components and the two project extensions. Blue and orange identify the model and training components of Qian et al.; green identifies the large-image density/colour method and the subsequent geospatial integration developed in this dissertation. The dashed link is the separate independent-mesh experiment [2].

The paper reports training on VIGOR-derived cross-view pairs, evaluation outside the training cities, and a large-scale strategy based on overlapping patches. Its analysis identifies pose accuracy, orthographic projection, locally flat ground, and atypical architecture as important reconstruction factors. The present study extends the systems context by examining local-to-world transforms, seam handling, DSM alignment, building association, and ChordAtlas publication [2].

Three processing domains must be distinguished. The released Sat3DGen large-image wrapper predicts overlapping density volumes, applies hard boundary cropping and equal-weight averaging, and runs Marching Cubes once on the fused field; unlike its single-image mesh path, the wrapper does not query or export vertex colour [2, 29]. The large-image extension developed here retains one global density-domain surface extraction but maps window locations at sub-voxel precision, feathers overlapping contributions, and performs a memory-bounded second pass for vertex RGB. The subsequent geospatial path operates in mesh space, adding OSM/DSM alignment, footprint association, per-

building extraction, and ChordAtlas publication.

2.3 Semantic city models, mesh validity, and provenance

A renderable triangle soup is not yet a semantic city model. CityGML 3.0 separates geo-referenced geometry from urban-object semantics, relationships, and multiple spatial representations [30]. Its Levels of Detail should not be reduced to a generic low/medium/high polygon-count switch: geometric complexity and semantic granularity are distinct [31]. The bridge’s footprint-linked identifiers are therefore a useful coarse association; the current OBJ outputs do not encode the validated roof and wall semantics required for a CityGML representation.

Downstream validity is similarly task-dependent. Local mesh repair may close holes or remove defects, but it can also change the observed surface [32]; standards-oriented validation distinguishes valid primitives from merely readable files [33]. This motivates reporting boundary and non-manifold edges separately from geometric accuracy.

Finally, a manifest is only a container unless it records causal lineage. PROV-O distinguishes entities, activities, and agents [34]; in this setting the equivalent minimum is an exact input allow-list and hashes, geographic extent and CRS, code/model/configuration identity and execution state.

2.4 Geospatial contracts

2.4.1 Coordinate order and reference systems

GeoJSON positions are written in longitude–latitude order [35], whereas an Overpass QL bounding box is ordered south, west, north, east [36]. Confusing the two conventions

can send a valid query to the wrong region without producing a type error. This is a representative integration defect: each subsystem may accept four floating point values while disagreeing about their meaning.

The data-builder grid uses Web Mercator world pixels at zoom 20. EPSG:3857 is appropriate for web mapping, but its registry explicitly notes scale distortion [37]. Metric analysis of the London DSM instead uses EPSG:27700, the British National Grid [38]. The bridge establishes a local frame whose origin is stored in WGS84 and whose axes are X east, Y height, and Z south. The frame is convenient for ChordAtlas and Sat3DGen mesh operations, but a constant-metres-per-degree approximation is not equivalent to a projected CRS. Chapter 4 quantifies this difference.

2.4.2 Surface elevation and semantics

The Environment Agency’s 1 m LiDAR Composite DSM records surface height rather than bare-earth terrain. It can therefore include buildings, vegetation, and vehicles; its 2022 composite also joins surveys acquired at different times [4]. GeoTIFF provides the georeferenced raster interchange contract [39]. A DSM can help place generated geometry vertically, but it is not an exact and contemporaneous building-height ground truth.

OSM supplies footprints and contextual categories. It is a collaborative database under the ODbL [3]. Overpass queries commonly return ways, relations, and the supporting nodes needed to reconstruct their geometry. Those support nodes must not automatically become category features, and multipolygon outer/inner rings must retain their roles. The data audit in Chapter 4 shows why semantic masks cannot be accepted solely because image files were produced.

2.5 Synthesis and research gap

The literature reviewed above addresses complementary parts of the target workflow. Cross-view models connect satellite imagery to density fields, rendered views, or explicit meshes [19, 2]; procedural systems support structured and editable urban content [5, 1, 7]; and OSM, DSM, and city-model standards provide geographic context, elevation, and semantic conventions [3, 4, 30]. These components use different spatial frames and representations, so their outputs do not compose automatically.

Among the reviewed approaches, none follows overlapping satellite-conditioned predictions through regional fusion, geospatial registration, footprint-based building extraction, and publication into an interactive procedural environment. Neural reconstruction work primarily evaluates geometry or rendering, whereas procedural workflows generally begin with geometry and geographic structure already available. The research gap is therefore the transition between these representations and the evaluation of that transition as an integrated workflow.

Accordingly, this study evaluates whether satellite tiles, projected DSM data, local mesh coordinates, and ChordAtlas placement remain spatially consistent. It measures the effects of fusion, cropping, stitching, vertical correction, and building extraction. These evaluation dimensions are operationalised in Chapter 3 and provide the basis for the research question introduced in Chapter 1.

3 | Methodology

3.1 Research scope and methodological rationale

The empirical design evaluates six linked components: London data preparation, large-image density and colour inference, independent-mesh processing, static and on-demand ChordAtlas integration, a matched representation comparison, and component verification. Together, they test whether overlapping satellite-conditioned predictions can form a continuous coloured surface and be converted into coordinate-explicit, building-level assets for an interactive procedural environment.

A contiguous Sat3DGen experiment measures overlap fusion and global colour recovery. A separate experiment uses retained OBJ outputs to isolate geospatial placement, mesh stitching, OSM/DSM processing, and building extraction. Both routes are evaluated at the ChordAtlas publication boundary. Data-quality tests characterise the London multimodal collection. Retained ChordAtlas appearance exports provide representation-level evidence.

3.2 Systems evaluation design

The evaluation combines data-integrity analysis, coordinate tests, staged geometry measurements, controlled regression cases, topological inspection, and cross-language integration tests. Source tracing establishes the transformations applied by the active pipeline; generated artefacts and manifests provide stage-level measurements; and executable checks test the coordinate, geometry, cache, and publication contracts. The ChordAtlas bridge is compared with the existing workflow at the level of representation, selection, conversion, and publication behaviour.

The research was organised around five evaluation questions:

- EQ1** Are the source data and pairing conventions internally consistent enough to support the intended cross-view and geospatial uses?
- EQ2** How do density-coordinate placement, overlap weighting, and second-pass colour recovery affect boundary continuity, colour completeness, and resource use in large-image inference?
- EQ3** How did the executed mesh-processing stages affect mesh structure, continuity, and height, and was their observed behaviour consistent with the intended design?
- EQ4** What do the static and on-demand paths demonstrably achieve in ChordAtlas, and where do their atomicity and cache boundaries lie?
- EQ5** How do the retained CityEngine and ChordAtlas artefacts differ in representation from the two matched Sat3DGen routes, and which comparisons remain valid without geometric or appearance ground truth?

3.3 Evaluated configuration

The evaluated configuration comprises the London data-preparation workflow, the released Sat3DGen checkpoint and large-image extension, the active independent-mesh package, and the ChordAtlas integration layer. Runtime imports and entry-point tracing identify the components used by each experiment. Functional analysis covers GUI entry points, worker and process boundaries, stable selection identities, manifest fields, cache validation, publication order, rollback, and validation depth. Reproduction details are recorded in [Appendix A](#).

3.4 System overview

Figure 3.1 presents the conceptual data flow derived from the active implementation. Chapter 4 assigns execution and validation status to the individual paths.

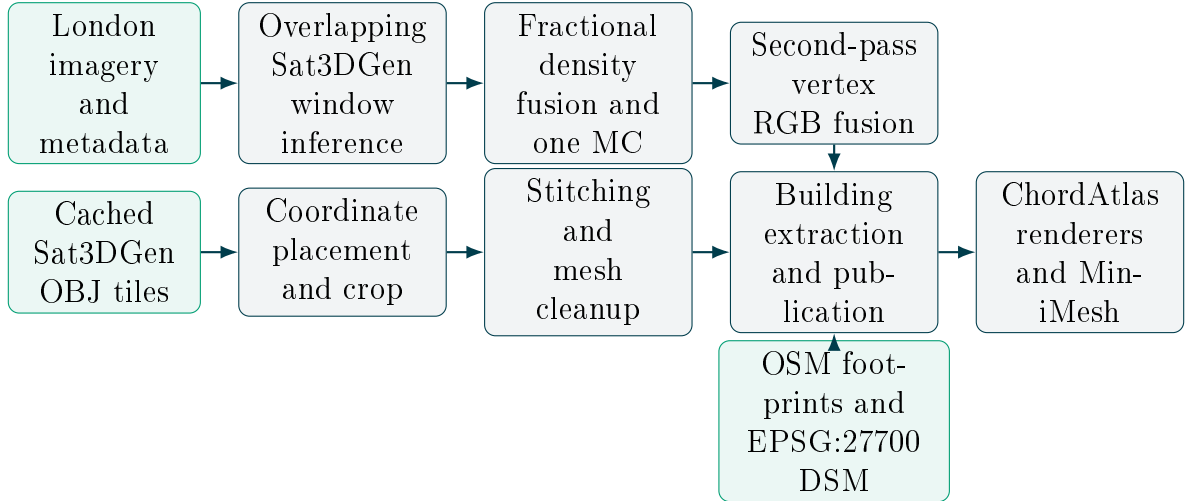


Figure 3.1: Conceptual project architecture. The upper route performs contiguous density-field fusion and colour recovery; the lower route evaluates independent-mesh assembly. Both converge on building extraction and publication before ChordAtlas. OSM footprints and the DSM enter from below as independent geospatial inputs rather than sequential processing stages.

3.5 London data preparation

The London data builder is evaluated as a staged data contract supporting cross-view pairing and downstream geospatial processing. The protocol measures coverage, duplication, partition leakage, label geometry, and auxiliary-channel consistency. Figure 3.2 links each processing stage to its available artefacts and audit denominator.

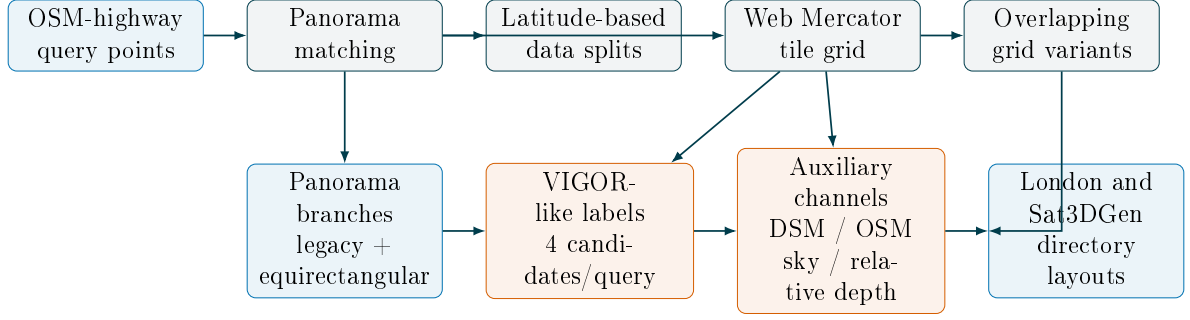


Figure 3.2: Data-builder stages and the contracts tested for completeness and geometric consistency.

3.5.1 Point matching, tiling, and imagery

The data-builder begins from WGS84 point features derived from an OSM highway query. A manually defined longitude–latitude region and a zoom-20 Web Mercator grid assign satellite centres. A panorama is matched when its point lies inside a satellite tile’s half-width and half-height. Chapter 4 reports the resulting population and coverage.

For longitude λ in degrees, latitude ϕ in radians, zoom z , and world-pixel scale $S = 256 \cdot 2^z$, the scripts use the standard Web Mercator form

$$p_x = S \frac{\lambda + 180}{360}, \quad (3.1)$$

$$p_y = S \left[\frac{1}{2} - \frac{1}{4\pi} \ln \left(\frac{1 + \sin \phi}{1 - \sin \phi} \right) \right]. \quad (3.2)$$

Latitude is clamped to the Web Mercator limit. The nominal ground resolution at latitude ϕ is

$$r(\phi, z) = \frac{2\pi R \cos \phi}{256 \cdot 2^z}, \quad (3.3)$$

where R is the Web Mercator sphere radius. At $\phi = 51.5074^\circ$ and $z = 20$, the implementation gives approximately 0.093 m px^{-1} , so a 640-pixel tile spans about 59.5 m. Values used in the audit were recalculated from the coordinate metadata.

The acquisition procedure requests 640-by-320 Street View images with a 120-degree field of view and 640-by-640 satellite images, nominally at zoom 20. The legacy branch does not retain the returned panorama identifier or snapped coordinates. The equirectangular branch requests six cube faces and projects them into 1,536-by-768 equirectangular panoramas. The projection and image dimensions were treated as data transformations, while observation independence was tested separately through content hashes.

3.5.2 Splits and candidate labels

The legacy training, validation, and test split sorts by latitude and assigns 70, 15, and 15 per cent respectively. Spatial sorting can reduce arbitrary mixing, but there is no boundary buffer, and duplicate returned panoramas can cross a boundary.

The VIGOR-style branch fills a 16-by-17 Cartesian grid of observed latitude and longitude centres and chooses the four nearest satellite centres for every panorama. For candidate j , an offset was classified as containing the query only when both absolute pixel components were at most 320. The audit recorded the count outside this square by candidate rank and the maximum absolute component. This tests geometric compatibility rather than mere four-column syntactic compatibility with VIGOR [18].

Two denser satellite grids were also reconstructed. A 320-pixel stride gives approximately 50 per cent overlap, while a 576-pixel stride gives 10 per cent overlap. Coverage calculations account for coordinate rounding and incomplete placement at non-divisible outer boundaries.

3.5.3 Auxiliary channels

The pipeline includes binary sky masks, model-derived relative depth, a DSM-derived normalised raster, and OSM semantic layers. The DSM branch transforms WGS84 coordinates into the raster CRS, reprojects a patch with bilinear sampling, and scales its 5th–95th percentile range independently. Consequently, it was evaluated as a normalised surface-height channel, not metric depth.

For OSM, raw element types and tags were counted separately from raster output. Supporting nodes returned by recursive Overpass queries were identified by their lack of category tags. Multipolygon reconstruction and inner-ring assignment were traced through geometry construction and drawing. This made it possible to distinguish an existing image from a semantically valid mask.

3.6 Coordinate and vertical-frame contract

The bridge defines a local frame at (λ_0, ϕ_0) using $M = 111,320$ metres per latitude degree:

$$x = (\lambda - \lambda_0)M \cos \phi_0, \tag{3.4}$$

$$z = -(\phi - \phi_0)M. \tag{3.5}$$

Thus X points east, Z points south, and Y is vertical. The inverse uses the same constants, so algebraic round trips should be exact to floating-point precision. The frame is illustrated in Figure [3.3](#).

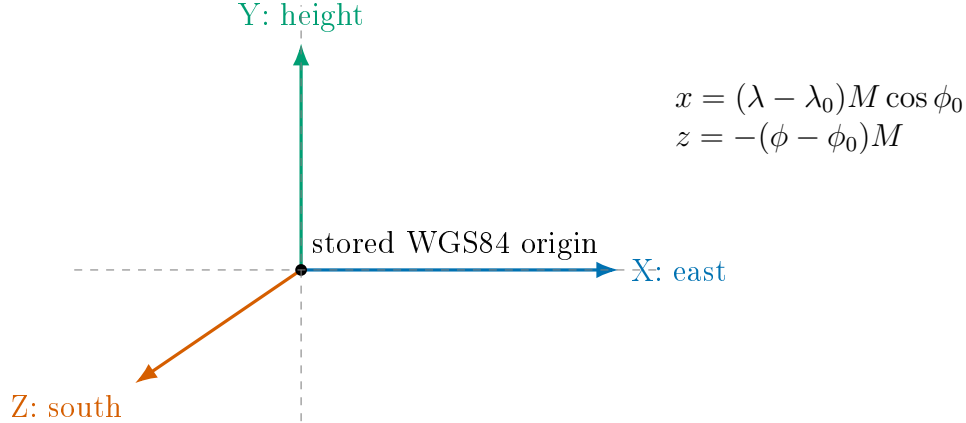


Figure 3.3: Local coordinate contract used by the bridge. The reversible local approximation is convenient for block-scale processing but is not identical to EPSG:27700.

Coordinate evaluation had two parts. First, the AOI centre and four corners were transformed to local coordinates and back, measuring residual in metres. Secondly, the same points were transformed from EPSG:4326 to EPSG:27700 using `pyproj`; projected easting/northing differences from the origin were compared with local X/south values. This separates inverse consistency from external metric agreement.

The vertical contract has three distinct quantities: generated-mesh local Y, DSM surface elevation, and a workspace vertical offset. Their values were read from manifests and paired OBJ vertices.

3.7 Large-image and mesh-processing protocols

3.7.1 Rectangular density-field fusion

The large-image extension retains the released Sat3DGen checkpoint and its standard 256-by-256 network transform. It accepts a rectangular source mosaic without first truncating it to a central square. In the evaluated setting, a 640-pixel source window moves by 160 pixels, giving 75 per cent nominal overlap; each source window is resized by the

unchanged model transform. The `mesh resolution` parameter is the volumetric sampling size, not the image input size. A value of $N = 192$ and the checkpoint's 0.8 position scale produce a cropped density patch of $154 \times 154 \times 173$ samples with a 19-voxel border.

Let s_i be the source-image start of window i on either horizontal axis and $L = 154$ the cropped density width. Its continuous density origin is

$$o_i = s_i \frac{L}{640}. \quad (3.6)$$

The origin is retained as a floating-point value rather than rounded to an integer cell. Each density sample is bilinearly distributed to the four neighbouring cells of the global horizontal grid. A separable raised-cosine window tapers the interior 19-voxel boundary of each patch; an edge coincident with the outer mosaic boundary retains unit weight. For a global density position p , the accumulated field is

$$\bar{D}(p) = \frac{\sum_i w_i(p) D_i(p)}{\sum_i w_i(p)}, \quad (3.7)$$

where D_i is the resampled prediction and w_i combines its bilinear placement and spatial taper. A zero denominator is treated as an invalid coverage state. Marching Cubes is applied once to $\bar{D}$ at the evaluated isovalue of 4.5, so overlap is resolved before surface extraction.

Algorithm 1 gives the corresponding semi-formal procedure. The two accumulators separate predicted density from its support weight, which permits a final normalised field to be extracted by a single Marching Cubes operation.

Algorithm 1 Fractional feather fusion of overlapping density predictions

Require: Rectangular mosaic I ; source-window starts S ; predictor M ; cropped density width L ; halo h ; isovalue τ **Ensure:** Global vertices V and faces F

```

1: Initialise density sum  $A \leftarrow 0$  and horizontal weight sum  $W \leftarrow 0$ 
2: for all  $(r_i, c_i) \in S$  do
3:    $D_i \leftarrow \text{PREDICTDENSITY}(M, I, r_i, c_i)$ 
4:    $D_i \leftarrow \text{CROPHALO}(D_i, h)$ 
5:    $(o_r, o_c) \leftarrow (r_i L / 640, c_i L / 640)$ 
6:    $K_i \leftarrow \text{RAISEDCOSINEWEIGHTS}(L, h, \text{mosaic-edge flags})$ 
7:    $(f_r, f_c) \leftarrow (o_r - \lfloor o_r \rfloor, o_c - \lfloor o_c \rfloor)$ 
8:   for all  $(a, b) \in \{0, 1\}^2$  do
9:      $\alpha_{ab} \leftarrow (af_r + (1 - a)(1 - f_r))(bf_c + (1 - b)(1 - f_c))$ 
10:     $\text{ACCUMULATE}(A, \alpha_{ab} K_i D_i, \lfloor o_r \rfloor + a, \lfloor o_c \rfloor + b)$ 
11:     $\text{ACCUMULATE}(W, \alpha_{ab} K_i, \lfloor o_r \rfloor + a, \lfloor o_c \rfloor + b)$ 
12:   end for
13: end for
14: require  $W(p) > 0$  for every covered horizontal position  $p$ 
15:  $\bar{D} \leftarrow A / W$   $\triangleright$  broadcast  $W$  over the vertical axis
16:  $(V, F) \leftarrow \text{MARCHINGCUBES}(\bar{D}, \tau)$ 
17: return  $(V, F)$ 

```

3.7.2 Memory-bounded vertex-colour recovery

The released large-image wrapper discards each tri-plane after density prediction and therefore exports geometry without the vertex-colour query used by single-image inference [29]. The added colour pass keeps the extracted global vertices and faces fixed. It reprocesses one source window at a time, reconstructs that window’s tri-plane, identifies the global vertices supported by the same fractional feather weights, and queries the pretrained colour network in batches.

The global Marching-Cubes axes are $[\text{row}, \text{column}, z]$. For a window with continuous origin (o_r, o_c) , a supported vertex is mapped back to the full 192^3 model grid as

$$q = [\text{column} - o_c + 19, \text{row} - o_r + 19, z + 19] \frac{2}{191} - 1. \quad (3.8)$$

The horizontal query is clamped only for the fractional halo outside the cropped density support; the support weight itself is evaluated without that clamp. With $c_i(v)$ denoting the RGB query for vertex v , colour is blended in floating point as

$$\bar{C}(v) = \frac{\sum_i w_i(v) c_i(v)}{\sum_i w_i(v)}. \quad (3.9)$$

A two-dimensional spatial index limits each window to nearby vertices, and only one tri-plane is resident at a time. Export requires positive colour weight for every vertex, finite RGB values, unchanged vertex and face arrays, and an exact colour round trip after reloading the PLY.

Algorithm 2 describes the second pass. It uses the geometry-fusion weights again, but accumulates RGB only for vertices whose continuous coordinates lie inside a window’s weighted support.

Algorithm 2 Memory-bounded recovery of colour on a fixed global mesh**Require:** Mosaic I ; fixed mesh (V, F) ; window plan S ; predictor M ; fusion metadata**Ensure:** Per-vertex RGB array C on unchanged (V, F)

```

1: Build a two-dimensional spatial index  $B$  over the horizontal axes of  $V$ 
2: Initialise RGB sum  $R \leftarrow 0$  and vertex weight sum  $Z \leftarrow 0$ 
3: for all windows  $i \in S$  do
4:    $T_i \leftarrow \text{ENCODEWINDOW}(M, I, i)$   $\triangleright$  one resident tri-plane
5:    $(J, w) \leftarrow \text{SUPPORTEDVERTICES}(V, B, i, \text{fusion metadata})$ 
6:   for all paired batches  $(Q, w_Q)$  drawn from  $(J, w)$  do
7:      $q \leftarrow \text{MODELCOORDINATES}(V[Q], i)$ 
8:      $c \leftarrow \text{QUERYCOLOUR}(M, T_i, q)$ 
9:      $R[Q] \leftarrow R[Q] + w_Q c$ 
10:     $Z[Q] \leftarrow Z[Q] + w_Q$ 
11:   end for
12:   Release  $T_i$ 
13: end for
14: require  $Z(v) > 0$  and finite for every  $v \in V$ 
15:  $C \leftarrow \text{QUANTISERGB}(R/Z)$ 
16:  $\text{EXPORTANDRELOADVERIFY}(V, F, C)$ 
17: return  $C$ 

```

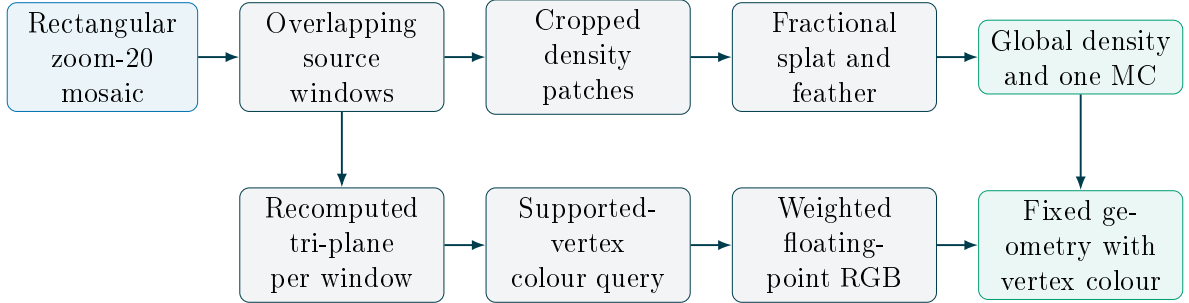


Figure 3.4: Two-pass large-image method. The first pass blends density before a single surface extraction; the second revisits each window to blend colour on the unchanged global vertices.

3.7.3 Independent-mesh stages and relation to Sat3DGen

The evaluated sequence loads configuration, OSM, and DSM data; discovers OBJ tiles; parses, crops, transforms, classifies, and pre-aligns each tile; removes lower geometry;

stitches vertex groups; applies DSM correction; and exports paired DSM and no-DSM building branches. The experiment isolates these post-inference operations from upstream generation and optional façade processing.

Table 3.1 distinguishes the released large-image wrapper, the density/colour extension, and the independent-mesh geospatial path.

Table 3.1: Relationship among the released Sat3DGen large-image wrapper, the density/colour extension, and mesh-space geospatial integration.

Aspect		Released wrapper	large-image	Density/colour extension	Mesh-space and ChordAtlas path
Learned representation		Released conditioned density field	DINOv3-tri-plane	Same checkpoint and model transform; no retraining	Released model or cached OBJ enters explicit local coordinates
Overlap		Integer-aligned fields and equal-weight averaging	cropped equal-weight	Fractional origins, bi-linear placement, and raised-cosine weights	Independent OBJ tiles are cropped, placed, and stitched
Surface colour	and	One global Cubes surface; large-image PLY has no colour	Marching large-image	One global surface followed by weighted per-vertex RGB recovery	RGB OBJ assets, building extraction, and a separate ChordAtlas colour renderer
Geospatial context	con-	Normalised scene coordinates	text	Rectangular mosaic and explicit image-to-density scale	X-east/Y-up/Z-south frame, OSM footprints, and EPSG:27700 DSM

Each stage was quantified using OBJ header counts and, where necessary, streamed vertex comparison. The following metrics were computed:

- vertices and faces before crop, after crop, after bottom removal, after stitching, and after building extraction;
- geographic cluster count and separation from filename coordinates;
- number and distribution of per-vertex DSM ΔY values;
- finite vertices, valid indices, zero-area and duplicate faces, boundary and non-manifold edges, connected components, and edge-incidence watertightness for a representative paired building;

3.7.4 Controlled tests of geometry rules

Two small synthetic fixtures tested whether documentation and configuration were enforced. For bottom removal, one triangle had only one vertex within 0.5 m of the global minimum and a second triangle was fully elevated. The stated all-three-vertices rule predicts that neither should be removed. The active function was invoked without modifying source code and its returned face set recorded.

For stitching, two identically wound triangles had coincident X/Z coordinates but were separated by 10 m in Y. They were assigned different tile ranges and stitched with an X/Z radius of 0.01 m. The experiment recorded vertex/face counts and retained attributes. It directly tests whether the configured maximum Y difference participates in matching.

3.8 ChordAtlas integration protocols

3.8.1 Static workspace

The static bridge combines a target bounding box, geospatial footprints, existing mesh tiles, panoramas, and DSM prerequisites in a staged workspace. Python records the manifest and local-frame footprint geometry; Java constructs the ChordAtlas object graph and converts OBJ geometry into indexed MiniMesh tiles. A stored source offset preserves the original centroid during conversion instead of implicitly recentering the input.

Validation checks required files, footprint geometry, MiniMesh structure, and coordinate metadata. Mesh coverage was independently intersected with the target AOI.

3.8.2 On-demand selection and publication

The on-demand path is embedded in the existing ChordAtlas GIS generator. A selected polygon is canonicalised by ring start, winding, holes, duplicate removal, and six-decimal

rounding before generating a stable selection identifier. Processing is serialised on one worker, and interface updates return to the GUI event thread.

The Java service writes an atomic request file and launches the Python bridge. The request distinguishes independent satellite-tile processing from the large-image mode. The tile path derives a geographically anchored allow-list and selects fresh or cached corrected geometry. The large-image path plans a rectangular zoom-20 mosaic, executes or validates the density and colour stages, and converts the fused mesh once into the project local frame. Both routes apply DSM processing and associate geometry with requested footprints.

Each publishable building receives an exact source-mesh manifest, geometry after ground processing, and metadata in a staging tree. Geometry ownership and filtering are applied per footprint rather than reusing one selection-wide mesh under several building identities. In large-image mode, Python additionally requires RGB on every OBJ vertex and verifies that DSM processing preserves colour indices. Java displays this OBJ through a dedicated vertex-colour renderer; the per-building MiniMesh is generated as a separate representation used by the ChordAtlas workspace. The staged Python publication is installed atomically within its own boundary and marked complete before Java conversion and renderer construction.

Figure [3.5](#) makes the transaction boundary explicit.

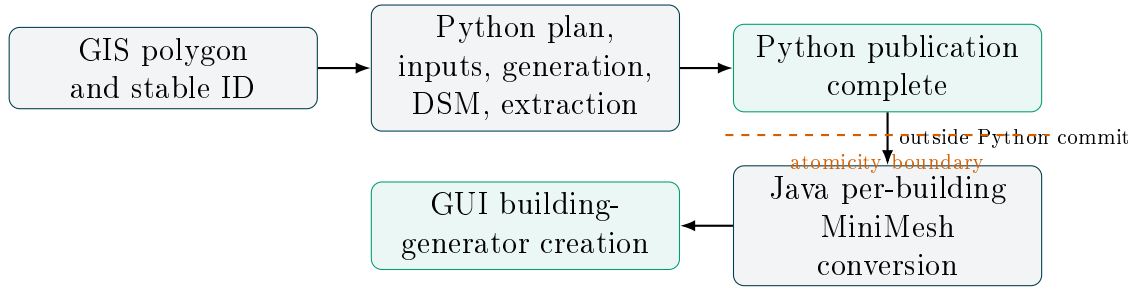


Figure 3.5: On-demand publication lifecycle. Python completion precedes the Java conversion required by ChordAtlas, so end-to-end publication is not a single atomic transaction.

3.8.3 Selection-scoped appearance inputs

The appearance branch reuses the same geographic selection and local frame as the building publication. For roofs, the bridge selects the satellite tiles already bound to the request, mosaics them in north-up orientation, clips them to the building footprint, and publishes a reference image and style mask. Existing ready references can be added to an earlier building publication without modifying its geometry. ChordAtlas validates the reference metadata before exposing it to the façade and roof style interface.

For street-level context, candidate viewpoints are sampled around the selected OSM footprint and converted to WGS84. Street View metadata are resolved and deduplicated before acquisition. The service presents one sample for approval, downloads the remaining accepted batch, promotes the complete set, and writes the panorama positions into ChordAtlas’s local coordinate frame. Figure 3.6 summarises the two paths.

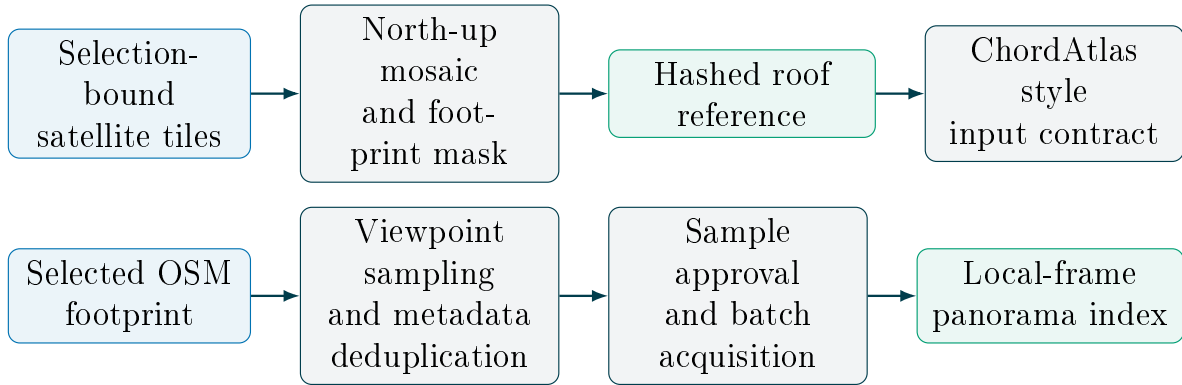


Figure 3.6: Selection-scoped appearance-input workflows. The roof path prepares a footprint-matched satellite reference for the style interface; the panorama path acquires and registers street-level observations in the same local frame.

Selection logs and result manifests were classified by processing state: request rejection, planning, acquisition, inference, corrected-scene reuse, building publication, Java conversion, and GUI creation.

3.9 Matched representation comparison

The directly matched comparison fixes OSM way 369245408 as the main-building identity. The large-image and independent-tile requests contain the same target bounds and the same two footprint geometries. The former begins from one fused density field; the latter begins from nine independently extracted meshes and adds OSM pre-alignment, lower-surface removal, stitching, and DSM processing. Building-level vertices, faces, relief, edge incidence, and vertex-colour variation are compared under this shared spatial request.

Cross-system inspection is restricted to representation. A retained CityEngine 2026 export records one named initial shape, UV coordinates, and UV-mapped materials. The two route-labelled ChordAtlas exports have envelopes consistent with the main footprint; the standalone ChordAtlas example has matched location. All examples are rendered with a common orthographic camera and metric scale within each figure. The

audit records polygon or triangle representation, vertex colour, texture coordinates, and diffuse/normal/specular map organisation. These properties are not treated as fidelity scores because no surveyed geometry or registered street-level appearance reference is available.

3.10 Verification protocol

3.10.1 Executable tests

Component tests cover configuration, coordinate frames, footprints, DSM prerequisites, OBJ inspection, tile planning, image integrity, manifests, cache freshness, publication, rollback, MiniMesh conversion, OBJ slicing, and vertex-colour parsing. Separate large-image regression cases exercise fractional placement, non-zero fusion coverage, seam continuity, colour-query coordinates, weighted RGB, and PLY round trips. The bridge suite also covers large-image selection, roof-reference and panorama planning, promotion, and coordinate transfer. Python, algorithm, and Java tests are reported separately because they exercise different runtime and data-transfer boundaries. Detailed results are provided in Appendix [A](#).

3.10.2 Validity review

Construct validity was assessed by matching each measure to the property it was intended to represent. The review distinguishes geographic extent from triangle coverage, file availability from semantic correctness, topological validity from geometric accuracy, and producer-side publication from consumer-side conversion. External validity is examined through the spatial and dataset coverage of the case study.

4 | Results and Analysis

4.1 Evaluation scope

Results are organised around the evaluation questions: data integrity, large-image density and colour inference, independent-mesh behaviour, coordinate control, static and on-demand integration, matched representation, and component verification. Quantitative findings are tied to the data population or processing stage from which they were measured.

4.2 Prototype feasibility and empirical focus

A retained London GLB and corresponding server record establish compatibility with the released Sat3DGen inference interface. The Unity/Ubiq client also compiled with the intended acquisition, request, retrieval, and scene-loading stages, informing the design of the downstream interface. The principal experiments comprise a fresh contiguous London large-image run, a matched overlap-fusion comparison, global vertex-colour recovery, an independent-OBJ mesh-processing study, and ChordAtlas publication. Retained ChordAtlas appearance exports provide representation-level evidence, while the earlier format-conversion prototypes contribute architectural context.

4.3 Data construction and integrity

The London builder is assessed for dataset readiness and downstream geospatial reuse. The results address coverage, independence, label geometry, and auxiliary-channel integrity rather than model-training performance.

4.3.1 Inventory and physical extent

Panorama acquisition covered the majority of target locations, while the satellite, DSM, OSM, and grid channels cover only a subset; sampling density and overlap also differ between branches.

Projected tile unions expose the difference between file count and physical coverage. The 127-tile set covers approximately 0.45 km^2 ; the repaired grid covers 0.96 km^2 . The 50 and 10 per cent overlap layouts cover approximately 0.93 km^2 and 0.89 km^2 , respectively. Despite containing 306 tiles, the latter supports 2,331/2,333 queries because its stride and boundary handling omit two locations. Tile count is therefore an insufficient coverage measure.

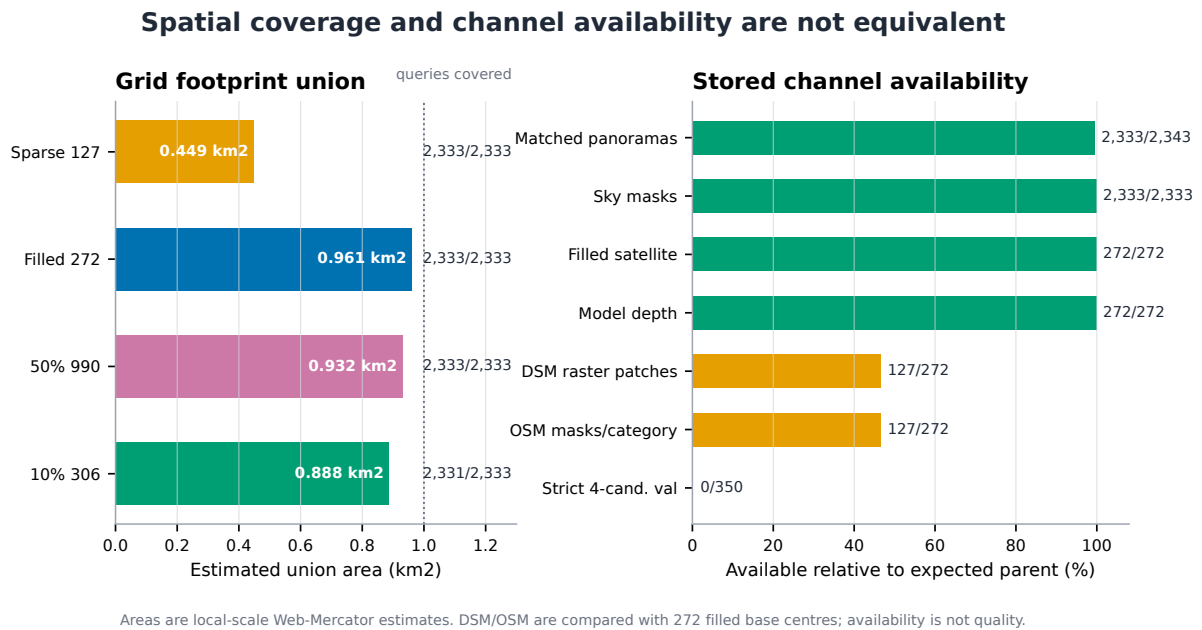


Figure 4.1: Estimated grid-footprint union, matched-query coverage, and stored channel availability. Areas use a local-scale Web-Mercator estimate.

4.3.2 DSM channel

The four Environment Agency inputs are adjacent 5,000-by-5,000 float rasters at 1 m resolution in EPSG:27700. The product is the Composite 2022 DSM [4]; it measures visible surfaces, not bare earth. The source rasters cover all 2,343 query points and all 272 repaired grid centres.

Figure 4.2 visualises the metric source mosaic before per-patch percentile normalisation. Mosaicking the four rasters across the approximately 1.00-by-0.94-kilometre footprint provides complete coverage at the displayed sampling grid.

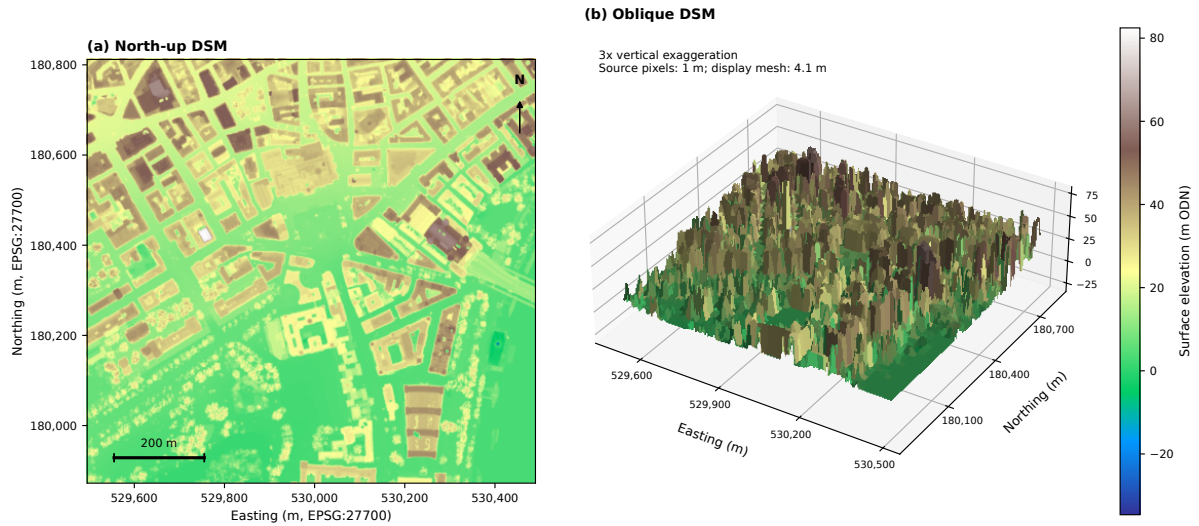


Figure 4.2: North-up and oblique views of the Environment Agency 2022 1 m first-return DSM mosaic over the London study area. The plan view preserves equal east–north scales and NoData; the oblique view uses threefold vertical exaggeration. Colour shows elevation relative to Ordnance Datum Newlyn. Source: Environment Agency, LIDAR Composite DSM 2022; Open Government Licence v3.0 [4, 40].

Independent 5th–95th-percentile scaling makes the patches suitable as relative surface-height channels but removes cross-patch comparability in absolute elevation. A legacy DSM derivative without traceable production metadata was therefore excluded from the evaluated channel.

4.3.3 OSM semantic channel

The Overpass query retrieves support nodes recursively, and the extractor emits all returned nodes as point features even when they have no category tags. Of 67,281 category records, 57,623 are such support-node records (85.65 per cent). Multipolygon fragments are also treated as independent outer rings, and inner-ring assignment can fill intended holes. Semantic validity therefore requires geometry- and tag-aware validation rather than file-count checks. The active mesh loader adds 19,682 ordinary-building and 9,237 height-building records to the same index [3].

4.4 Large-image density and colour inference

4.4.1 Contiguous London configuration

The contiguous run covered approximately 285.5 m by 475.8 m from a 3,072-by-5,120-pixel zoom-20 mosaic. Each 640-pixel source window was passed through the unchanged 256-pixel Sat3DGen image transform. A 160-pixel stride produced a 17-by-29 schedule, or 493 windows, with 75 per cent nominal overlap. The mesh-resolution setting was 192 voxels; it did not change the network image size. After fractional placement, the fused density volume measured $1232 \times 740 \times 173$ and had no zero-weight cells.

One global Marching Cubes extraction produced 3,122,922 vertices and 6,241,862 faces. Density inference and fusion required 129.0 seconds and reached 6.62 GiB peak allocated GPU memory.

4.4.2 Window-boundary comparison

The hard-box and fractional-feather runs used the same source image, checkpoint, windows, overlap, voxel resolution, and isovalue. The baseline is described as release-style

because it retains the released wrapper’s hard boundary crop and equal-weight overlap rule, while using the matched project settings rather than the release defaults. Figure 4.3 shows the resulting top surfaces.

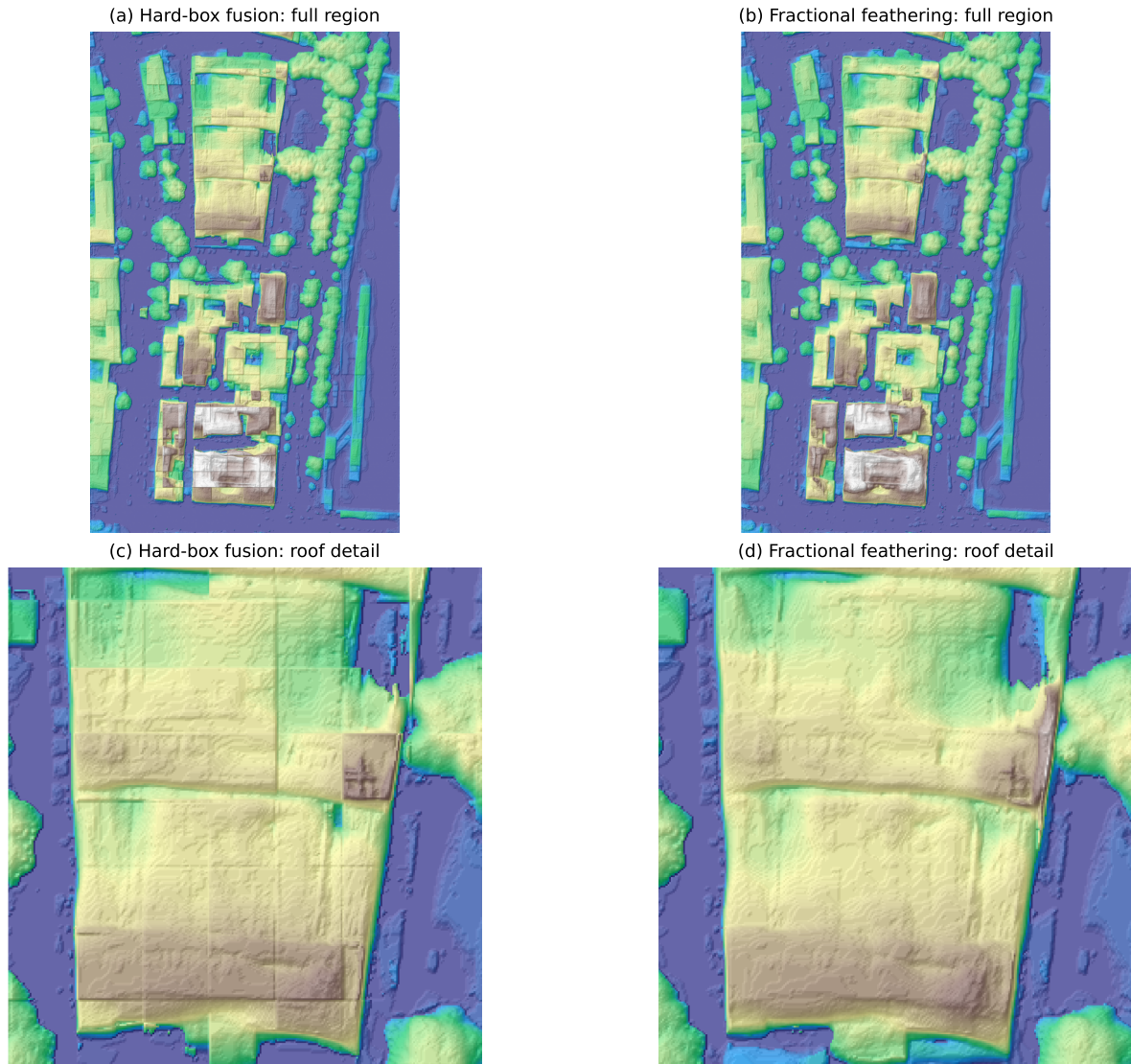


Figure 4.3: Matched large-image fusion comparison. Hard-box averaging produces repeated window-aligned height steps; fractional placement and raised-cosine feathering replace the abrupt contributor changes with smooth overlap transitions. The false colour and shading encode density-derived top-surface height, not satellite texture or appearance accuracy.

At the former hard-crop contribution boundaries, mean absolute density jumps fell from 0.710 to 0.286 along rows and from 0.963 to 0.491 along columns (Table 4.1). Mean top-surface jumps fell from 2.596 to 0.659 voxels and from 3.065 to 1.214 voxels, respectively. The corresponding reductions are 49–60 per cent for density and 60–75 per cent for the derived top surface. Geometry runtime increased by 8.6 per cent, while peak allocated GPU memory was unchanged at the reported precision. The comparison therefore supports a reduction in boundary discontinuity.

Table 4.1: Matched hard-box and fractional-feather fusion results.

Measure	Hard-box	Fractional feather	Change
Mean density jump, row boundary	0.710	0.286	−59.8%
Mean density jump, column boundary	0.963	0.491	−49.0%
Mean top-surface jump, row (voxels)	2.596	0.659	−74.6%
Mean top-surface jump, column (voxels)	3.065	1.214	−60.4%
Geometry stage (s)	118.7	129.0	+8.6%
Peak allocated GPU memory (GiB)	6.62	6.62	unchanged

4.4.3 Global vertex-colour recovery

The second pass accumulated 42.6 million positive window contributions over the fixed global surface. Every vertex received support from at least one window, with no uncovered or all-black vertices. Reloading the coloured PLY preserved every vertex position and face index and reproduced the exported RGB values. The colour stage required 97.3 seconds, of which 80.9 seconds were model queries, and reached 1.86 GiB peak allocated GPU memory. The exported PLY displayed its blended vertex colours in MeshLab. These checks establish complete and index-consistent appearance transfer.

4.4.4 Current myProject large-image publication

The large-image path was also exercised through the ChordAtlas bridge on a 1,536-by-1,536-pixel, 49-window selection. Its retained job records a 340,825-vertex, 678,726-face

density surface followed by complete colour support. The geometry stage required 17.0 seconds and the colour stage 23.2 seconds. The final colour-enabled publication reused the validated geometry stage from the same retained job before running and validating the second pass.

After one local-coordinate conversion, the context crop contained 66,666 vertices and 132,440 faces. DSM processing covered every retained vertex and preserved its RGB index while changing vertical position by at most 30.7 m. Footprint extraction published two non-exclusive coarse building assets. Their geometric union, represented by the top-level selection crop, contains 17,571 vertices and 34,383 faces; the two building files share 266 vertex positions and 452 triangles. Both OBJ files contain RGB on every vertex and pass the Java colour parser; per-building MiniMesh trees also exist. In ChordAtlas, the RGB OBJ is displayed by a dedicated vertex-colour renderer, while MiniMesh remains a separate geometric workspace representation.

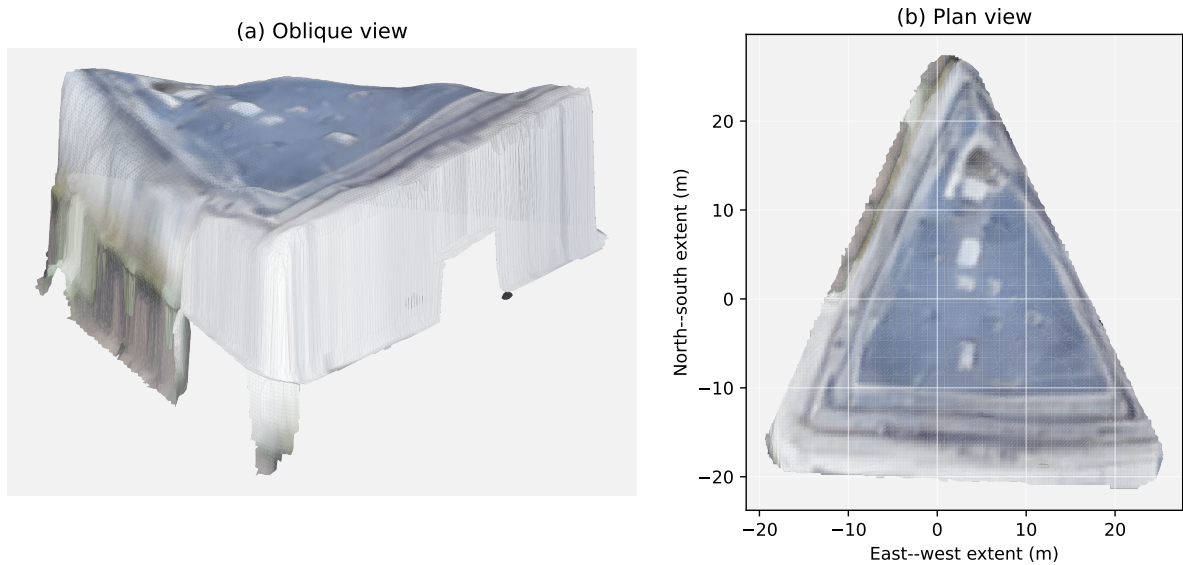


Figure 4.4: Oblique and plan views of the current large-image building publication after coordinate conversion, DSM processing, and footprint extraction. Face colour is the mean of the predicted per-vertex RGB values; the figure demonstrates colour retention through publication, not comparison with a reference texture.

4.5 Mesh-space post-inference experiment

4.5.1 Evaluated processing scope

The experiment processes 15 coordinate-paired OBJ tiles through OSM/DSM loading, crop and placement, lower-surface removal, stitching, DSM correction, and building extraction. The associated cache records describe geometry but do not bind a model checkpoint and configuration, so the measurements below characterise the mesh-processing stages.

Qian et al. report FID 19.2 and DSM RMSE 5.20 m on VIGOR-OOD [2]. Those metrics assess the published model protocol; this experiment measures downstream operations on retained mesh tiles and uses the external DSM as a processing input. Unlike the density-domain experiment in Section 4.4, it combines independently extracted meshes and therefore evaluates a complementary assembly strategy.

4.5.2 Geographic composition of the mesh batch

Recursive discovery selected two disconnected clusters: nine eastern tiles in a 3-by-3 arrangement and six western tiles in a 2-by-3 arrangement. Their centres are approximately 666 m apart, more than eleven nominal tile widths. Figure 4.5 plots every centre and nominal footprint.

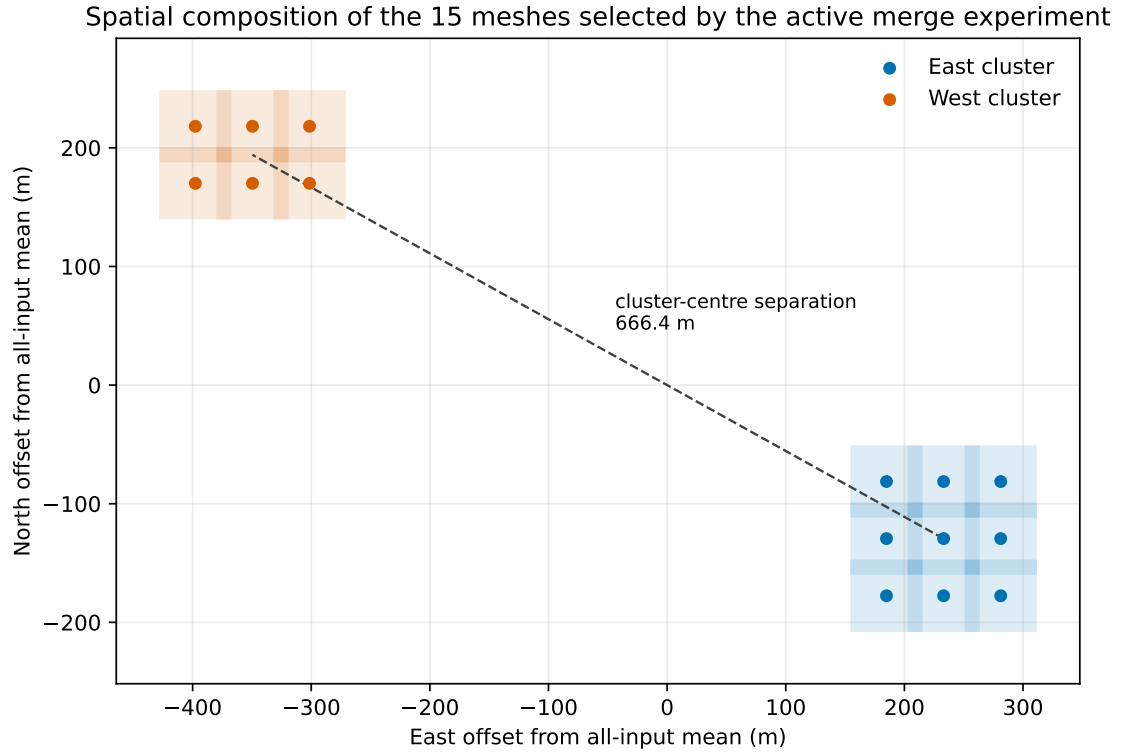


Figure 4.5: Geographic composition of the 15 OBJ meshes selected by the merge experiment. Filename coordinates identify two disconnected sites.

The merged output consequently combines two sites rather than one contiguous London scene. This geographic inconsistency originates in cache selection, not the upstream Sat3DGen generator [2].

4.5.3 Geometry retained through processing

Table 4.2 gives the exact geometry counts. From raw inputs to the stitched scene, 1,724,759/3,640,214 vertices (47.38 per cent) and 3,184,234/7,281,608 faces (43.73 per cent) were removed or consolidated. This large, intended reduction makes the crop, bottom-removal, and stitching predicates central to output validity.

Table 4.2: Geometry counts by post-processing stage.

Stage	Vertices	Faces	Change from previous stage
Raw 15 OBJ inputs	3,640,214	7,281,608	—
Cropped	2,828,613	5,615,039	−811,601 V; −1,666,569 F
Bottom removed	2,252,373	4,474,289	−576,240 V; −1,140,750 F
Stitched	1,915,455	4,097,374	−336,918 V; −376,915 F
DSM scene	1,915,455	4,097,374	topology unchanged
DSM building extraction	1,001,507	2,116,136	branch result
No-DSM building extraction	907,370	1,911,272	branch result

Figure 4.6 shows the saved crop-stage and bottom-removal outputs for one representative tile. The horizontal footprint is unchanged, while the operation removes a near-horizontal generated support sheet spanning almost the full tile. Every tile lost exactly 38,416 vertices and 76,050 faces at this stage, consistent with removal of the same standard lower grid.

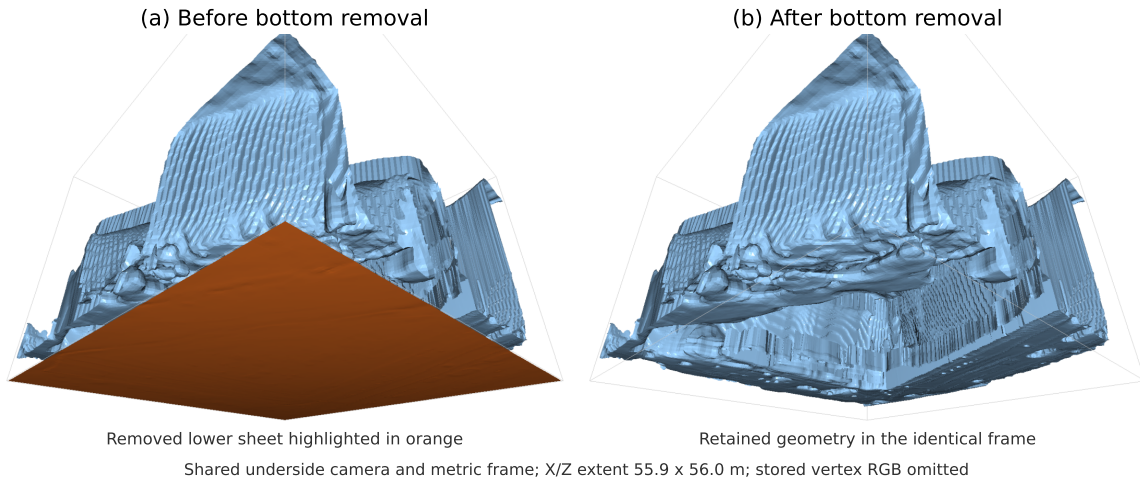


Figure 4.6: Representative tile 09 at two consecutive saved stages. Panel (a) is the cropped, placed, and OSM-pre-aligned mesh before bottom removal, with the removed lower sheet highlighted in orange; panel (b) is the retained mesh immediately afterwards. Both use the same underside camera and metric frame, and stored vertex colour is omitted to expose the geometric change.

The controlled bottom-removal fixture also showed that the active predicate removes a

triangle when only one vertex is close to the lowest surface, although the documented rule requires all three vertices to satisfy the threshold. In the stitching fixture, two triangles separated by 10 m in Y collapsed from six vertices to three because matching used X/Z coincidence without the configured vertical-separation test. In the north-western two-by-three subset used below, the six pre-stitch tiles contain 819,191 vertices and 1,626,029 faces; the saved post-stitch subset contains 696,199 vertices and 1,485,998 faces with the same horizontal extent. The sub-metre consolidation is quantified here because it is not distinguishable in a scene-scale silhouette.

4.5.4 DSM displacement is scene-wide

DSM correction changed all 1,915,455 vertices in the saved scene and every change was positive. The median displacement is 36.08 m, the mean is 42.24 m, and values range from 15.81 m to 62.63 m. The scene’s vertical range changes from -26.77 m to 18.76 m to 1.86 m to 67.94 m (Figure 4.7).

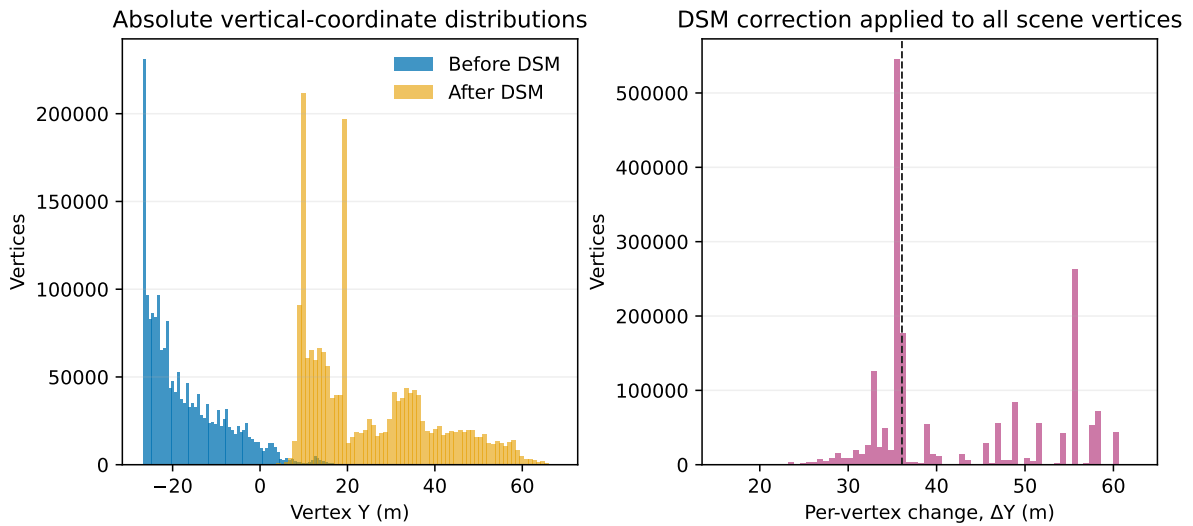


Figure 4.7: Per-vertex vertical displacement introduced by DSM correction. The distribution characterises implementation behaviour rather than surveyed height error.

4.5.5 North-western stitched scene and representative topology

Figure 4.8 isolates the north-western two-by-three cluster in the stitched scene before and after the saved DSM stage. The two OBJ files preserve vertex order, horizontal coordinates, RGB attributes, and face indices within this subset, so the stage changes only Y. The subset contains 696,199 vertices and 1,485,998 faces. All paired vertices move upward; median displacement is 38.58 m, with a 5th–95th percentile range of 32.96 m to 60.49 m.

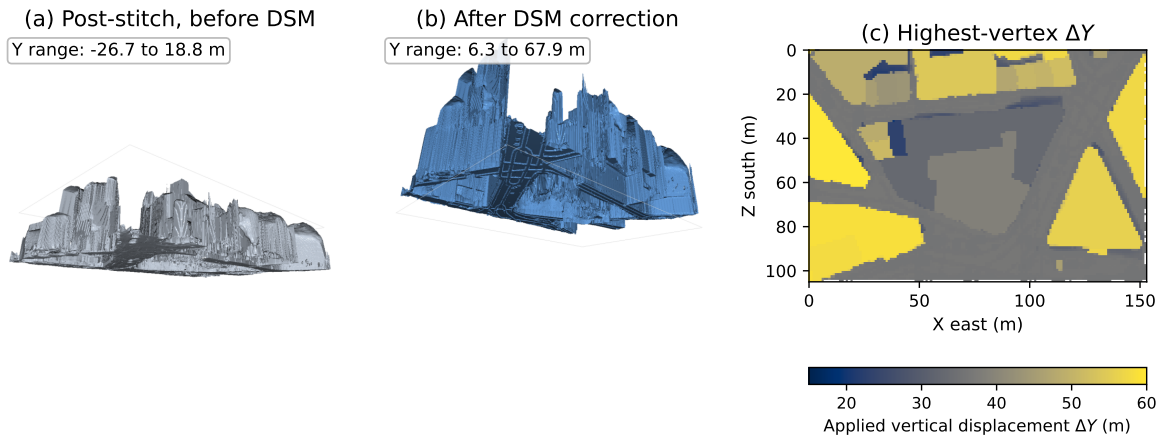


Figure 4.8: DSM-stage transformation in the north-western two-by-three stitched scene cluster. Panel (a) is the post-stitch output before DSM processing; panel (b) is the same mesh after DSM height correction. Both use uniform materials, a shared camera, and a common metric frame. Panel (c) reports paired ΔY at the highest pre-correction vertex in each occupied 1-metre X/Z cell. Stored RGB is omitted to expose the vertical change; the values describe implemented behaviour rather than surveyed elevation error.

A separate representative paired building remains a single connected component with finite vertices, valid indices, and no detected zero-area, duplicate, or non-manifold faces. Its mean paired vertical displacement is approximately 58.0 m. This building-level case establishes the presence of open boundaries; the population analysis in Section 4.7 evaluates their prevalence.

4.6 Static ChordAtlas integration

The static path converted a high-resolution source mesh into a structurally valid MiniMesh workspace with the required indices and generator structure. The mesh’s rectangular X/Z extent overlaps approximately 0.0096 km^2 of a 1.00 km^2 target, or 0.96 per cent. This is an axis-aligned extent proxy, not surface area, so it can include empty regions. Footprint-associated assets were exported, with overlapping OSM filter categories reported separately.

The evaluated result demonstrates

cached meshes $\rightarrow$ coordinate/footprint processing $\rightarrow$ MiniMesh conversion $\rightarrow$
structurally valid workspace,

using retained inputs. It establishes coordinate, footprint, format, and workspace integration over the available geographic extent.

4.7 On-demand ChordAtlas integration

4.7.1 Independent-tile publication population and asset quality

The tile-based on-demand set contains ten publication records with building-level metrics and one earlier record without them. Table [4.3](#) aggregates these records by corrected-scene decision.

Table 4.3: Independent-tile publication records by corrected-scene decision.

Record category	Publications	Tile references	Buildings	Publication evidence
Fresh correction	5	90	11	Modular building as-sets published
Reused correction	5	91	8	Modular building as-sets published
Earlier schema	1	9	—	Publication boundary reached; no current metrics

Across the ten metric-bearing independent-tile records, 181 tile references produced 19 building assets: eleven following fresh correction and eight following cache reuse. Figure 4.9 distinguishes selection size from the number of modular outputs.

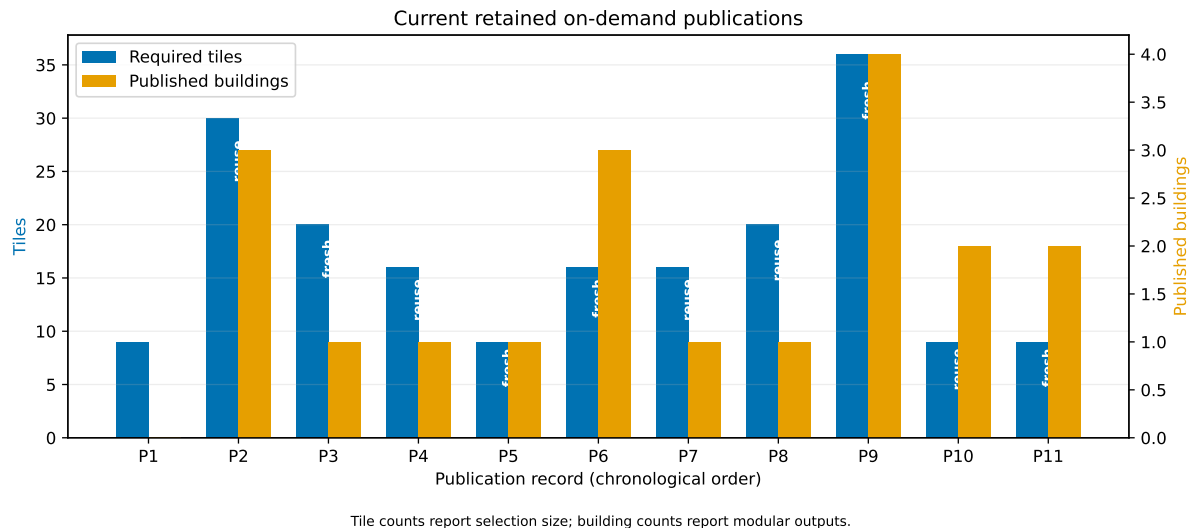


Figure 4.9: On-demand publication records, tile counts, building counts, and corrected-scene reuse decisions. The Python publication boundary precedes Java conversion.

Topology is the principal quality constraint in this published population.

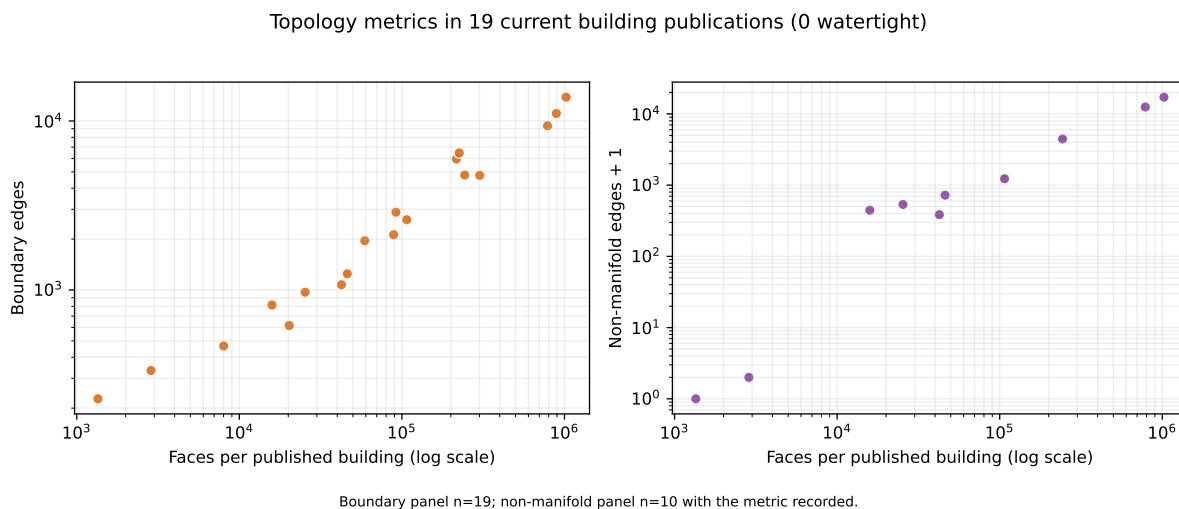
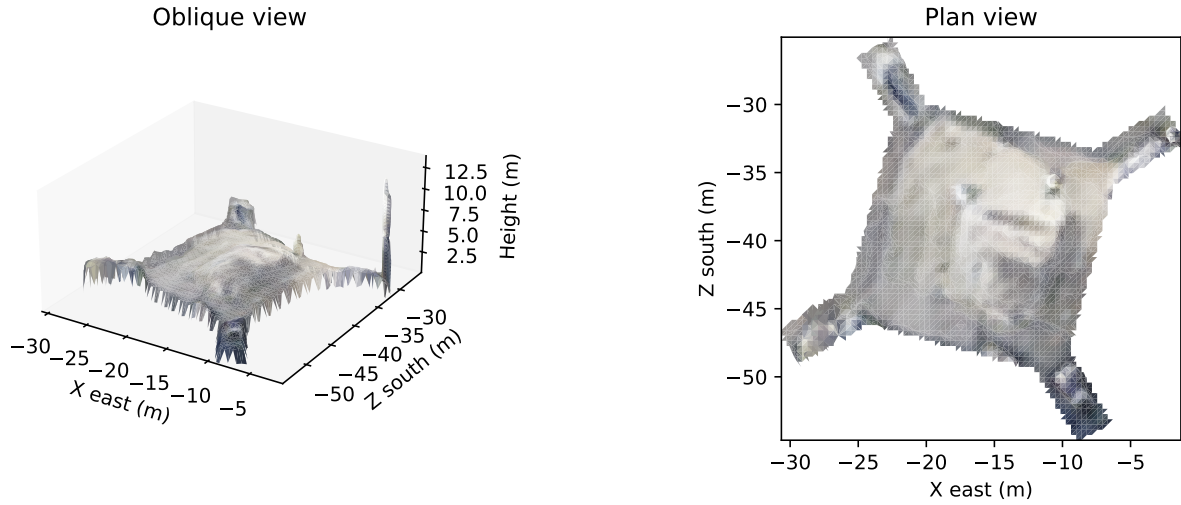


Figure 4.10: Boundary-edge metrics for all 19 independent-tile building publications and non-manifold-edge metrics for the ten v2 publications that record this field.

RGB attributes are preserved throughout the independent-tile publication set: every vertex in all 19 cropped OBJ assets carries colour data. The representative output in Figure 4.11 shows this appearance together with the open-edge topology identified above.

All 19 independent-tile buildings also contain MiniMesh indices. A real-asset Java check parsed vertex colours and constructed the generator layer successfully, confirming appearance transfer through OBJ parsing and generator construction.

Real per-footprint vertex-colour OBJ (4,192 vertices; 8,012 faces)



Project output, status COARSE_READY; appearance is not a ground-truth accuracy assessment.

Figure 4.11: Oblique and plan views of a representative building-level vertex-colour OBJ. Colour visualises the published appearance attributes.

4.7.2 Spatially matched appearance references

The independent-tile bridge extends building publication with two forms of appearance input. Four building assets have ready satellite roof references generated from the tiles bound to their selections. Each reference covers its source footprint completely, records a footprint-derived style mask, and leaves the published geometry unchanged. This creates a spatially matched roof-appearance input for the ChordAtlas style interface.

A selected-block panorama run also completed its planning, sample approval, batch acquisition, and promotion stages. Eight unique Street View panoramas were acquired and indexed in the ChordAtlas workspace using the project's local coordinate frame. The result connects a chosen OSM footprint to a deduplicated, selection-specific set of street-level observations.

Together, these results extend the bridge from geometry delivery to registered appear-

ance inputs. For this reference-registration experiment, the roof-reference style contract and panorama services pass their component checks; the evaluated outcome is reference preparation and spatial registration rather than a texture-quality measure.

4.7.3 Failure containment

Input validation rejected a duplicate footprint identity and an incomplete imagery set before inference. Both cases reached explicit terminal states, demonstrating fail-closed behaviour at the request and input-validation boundaries. Append-only selection logs preserve these transitions.

4.8 Matched London building comparison

4.8.1 Directly matched Sat3DGen routes

The large-image and independent-tile manifests request the same target bounds and the same two OSM footprints. Figure 4.12 compares the main building, OSM way 369245408, with identical cameras and metric display scale. Both outputs store per-vertex RGB produced within their respective Sat3DGen routes, and both were processed against the same one-metre DSM with complete recorded vertex coverage. Table 4.4 reports corresponding geometry, topology, relief, and colour-variation descriptors.

Table 4.4: Descriptive metrics for the directly matched main-building outputs.

Route	Vertices / faces	Relief (m)	Boundary / non- manifold	Luminance s.d.	Mean edge RGB Δ
Large-image	16,644 / 32,597	28.12	681 / 0	0.154	0.053
Independent tiles	21,555 / 42,524	38.07	1,074 / 384	0.213	0.082

The large-image sample has a more compact surface and fewer recorded boundary and

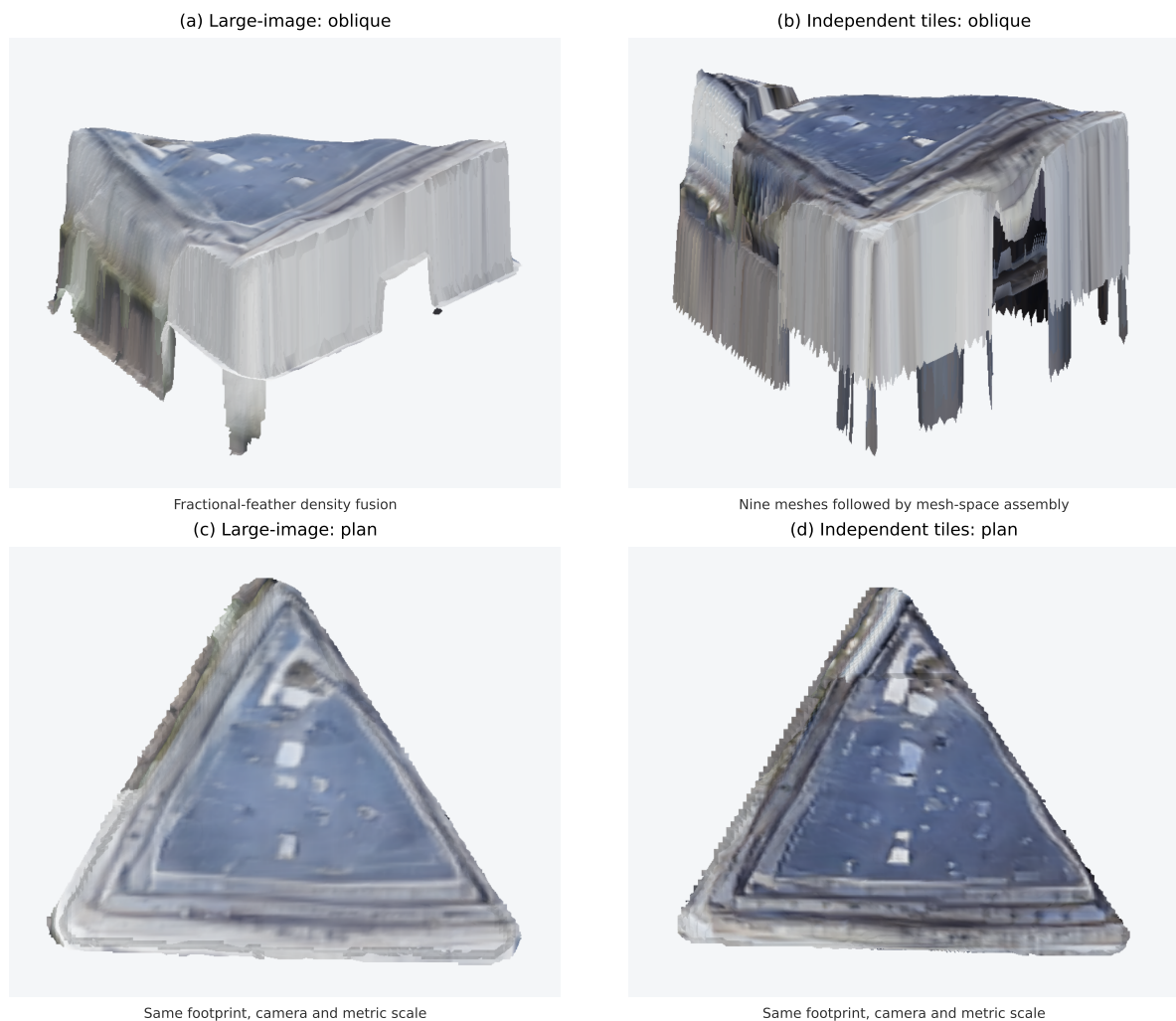


Figure 4.12: Directly matched building comparison for OSM way 369245408. Panels (a,c) show the fractional-feather large-image publication and panels (b,d) the nine-mesh independent-tile publication. The latter retains additional sheets, spikes, and detached fragments in this sample. Colours are per-vertex RGB retained from the two Sat3DGen routes, not satellite pixels projected directly onto the mesh.

non-manifold edges. The independent-tile sample contains more geometry and stronger local RGB variation. The latter observation is consistent with visibly higher-frequency colour detail, but it is not evidence of more accurate colour. Similarly, complete DSM coverage shows that both outputs were vertically conditioned.

4.8.2 Procedural appearance representations

Figure 4.13 places the CityEngine export and retained ChordAtlas artefacts under a common presentation camera and metric scale. These are illustrative representation examples rather than a common reconstruction benchmark. The route-labelled ChordAtlas envelopes are geometrically consistent with the main footprint, but their OBJ files do not embed its OSM identifier. The standalone ChordAtlas model has neither a matched location nor a retained run manifest.

The CityEngine 2026 log records one initial shape exported as 120 vertex records, 49 polygon records, 84 texture coordinates, and three materials. Its facades are regular, and the visible doors and windows reside in repeated texture atlases rather than separate door or window geometry. This supports a conclusion about export compactness and material organisation, not about site-specific texture accuracy [7, 17].

The two ChordAtlas exports contain no vertex RGB. Instead, the large-image route stores 27 and the independent-tile route 47 complete diffuse, normal, and specular map sets. Window-labelled materials and the aligned map triplets provide a multi-channel material representation rather than a single RGB value at each vertex. Figure 4.14 illustrates that distinction. The tile-derived export also contains one material named `error` on ten face records.

The retained route-labelled artefacts place satellite-route vertex colour and structured procedural appearance maps side by side, although the OBJ files do not provenance-lock a direct conversion between them. They do not show that ChordAtlas preserves

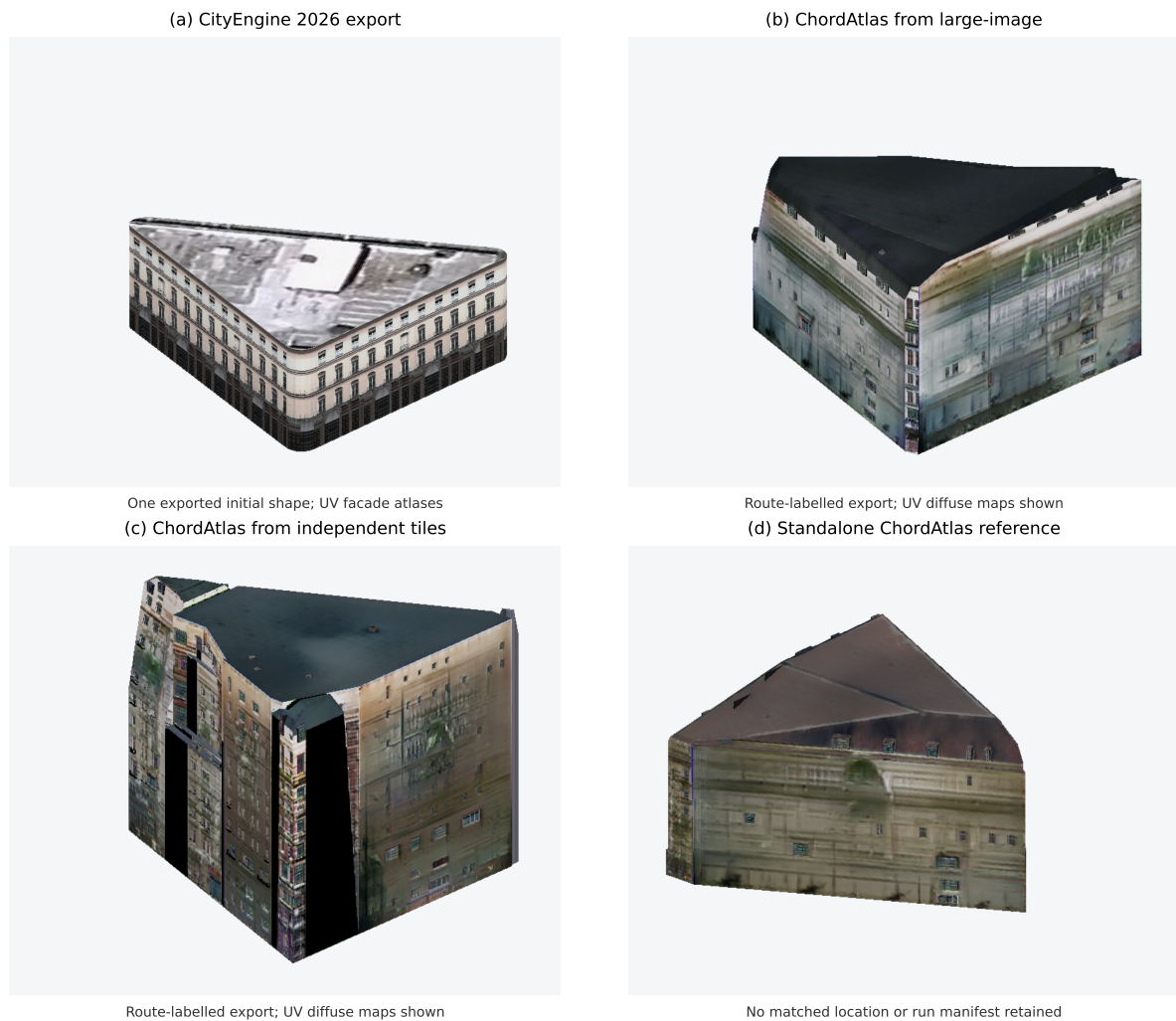


Figure 4.13: Illustrative procedural representation examples, centred without rescaling. The CityEngine export is a compact UV-mapped envelope with repeated facade atlases. The two route-labelled ChordAtlas exports use generated UV materials rather than vertex-RGB encoding and preserve the broad building envelope. The standalone ChordAtlas asset is included only as a stored contextual reference. Different or undocumented inputs prevent a fidelity ranking across panels.

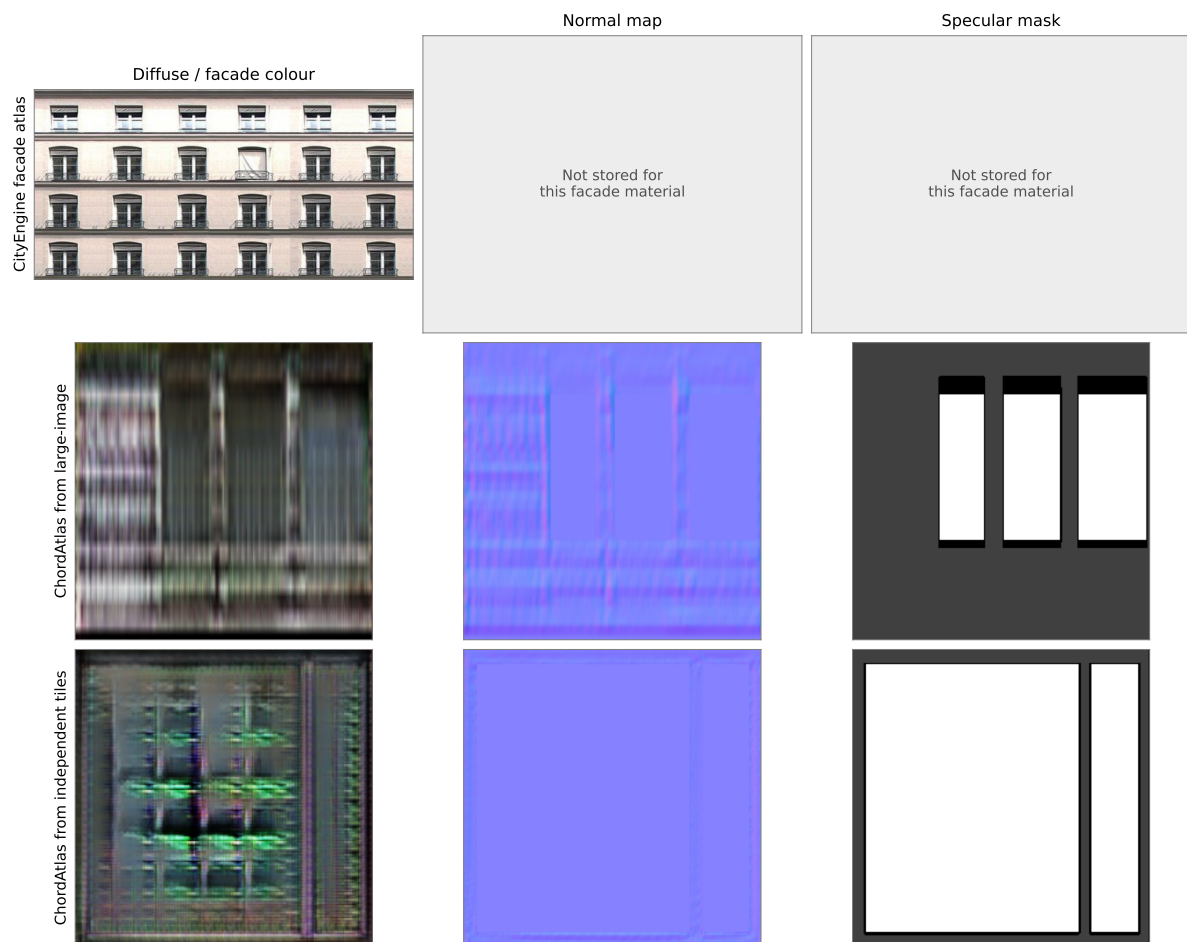


Figure 4.14: Selected facade material encodings. The CityEngine facade material uses a repeated diffuse atlas; no normal or specular map is stored for that facade material. The retained ChordAtlas procedural appearance exports store aligned diffuse, normal, and specular maps for window-labelled regions [6, 1]. These maps can depict frames, panes, and material response.

the Sat3DGen triangulation, reconstructs the true facade texture, or produces a more accurate building.

4.9 Results synthesis

All component tests passed within their specified scope, supporting the coordinate, cache, geometry-processing, and publication contracts across Python and Java. Appendix [A](#) provides the detailed test categories and results. Tables [4.5](#) and [4.6](#) consolidate the principal findings by evaluation dimension.

Table 4.5: Synthesis of principal findings and their evidential scope.

Evaluation dimension	Evidence basis	Finding and interpretation
Static ChordAtlas conversion	Structural inspection	A cached high-resolution mesh produced a valid MiniMesh workspace.
London data integrity	Hash and spatial analysis	Exact panorama duplication, split leakage, candidate-offset errors, and OSM support-node contamination require correction before learning use.
Large-image reconstruction	Matched run and round-trip inspection	Feathering reduced window-boundary discontinuities; every global-mesh vertex received colour without changing geometry, and the current ChordAtlas path retained RGB through publication.
Matched building representation	Shared OSM request, OBJ/MTL audit, and common-camera rendering	The large-image sample is more compact, while the tile sample has stronger colour variation and more fragments; CityEngine and ChordAtlas differences support representation-level, not accuracy, conclusions.
Building and appearance publication	Execution and component tests	Nineteen independent-tile assets and two current large-image assets were published; the paths retain RGB and provide either registered appearance references or the second-pass colour renderer.

Table 4.6: Synthesis of coordinate and geometry-reliability findings.

Evaluation dimension	Evidence basis	Finding and interpretation
Coordinate control	Calculation and projection comparison	The local transform is reversible, with up to approximately 19.1 m disagreement relative to EPSG:27700 displacement.
Stitching rule	Controlled regression	A 10 m-separated fixture collapsed from six to three vertices because vertical separation was omitted.
DSM correction rule	Artefact comparison	Every scene vertex moved positively.
Building topology	Population inspection	All 19 independent-tile publications and both current large-image publications retain boundary edges and none is watertight; nine of the ten v2 independent-tile assets reporting non-manifold incidence contain such edges.

5.1 Interpretation of the research scope

This dissertation examines how overlapping satellite-conditioned predictions can form a continuous coloured regional surface and then become editable urban assets. It contributes at two interfaces: density-domain fusion and colour recovery around the released Sat3DGen model, followed by geospatial association and publication through OSM, DSM, and ChordAtlas. This addresses the gap between plausible window-level generation and spatially identified, reusable urban content.

5.2 Research question: extent and reliability boundary

The study establishes a working large-image reconstruction and coordinate-explicit integration at building-asset scale. Overlapping density windows are fused into one surface, per-vertex appearance is recovered in a second pass, and the resulting mesh can be transformed, associated with OSM footprints, processed against DSM context, and converted for ChordAtlas. Stable selection identities, manifests, cache checks, and validation at the Python–Java boundary make these transformations inspectable rather than implicit.

Reliable regional operation depends on four properties highlighted by the results. Overlapping predictions need continuous placement and smooth weighting; spatial predicates must prevent unrelated or vertically separated surfaces from being combined; geometry processing must preserve topology and apply external height information selectively; and publication must be acknowledged by the consuming application. These requirements connect large-area reconstruction quality to asset-level system reliability.

5.3 Contribution of the ChordAtlas integration

The contribution is defined by the new data and execution contracts introduced at the ChordAtlas boundary. The integration extends the existing GIS workflow in the following ways:

- a coordinate contract that preserves geographic selection, a local origin, and building identity across mesh conversion;
- a selectable large-image source mode that binds the mosaic plan, density fusion, colour recovery, coordinate conversion, and footprint crop;
- manifest-driven planning and cache validation that bind a request to its required tiles and processing stages;
- OSM- and DSM-aware processing that associates generated surfaces with footprints and publishes separately editable building assets;
- selection-scoped roof references and panoramas, exercised for the independent-tile selections, that connect appearance inputs to the same footprint identity and local coordinate frame;
- staged Python–Java publication with explicit failure states, background execution, and a dedicated per-vertex-colour renderer; and
- executable checks for coordinate conversion, mesh prerequisites, selection manifests, cache state, and publication contracts.

Together, these changes turn a fused or independently assembled surface into geographically identified ChordAtlas assets whose processing history and failure state can be examined. Sat3DGen provides learned geometry and appearance, while ChordAtlas provides asset structure and procedural editing. The bridge preserves the coloured OBJ for display and builds MiniMesh as a separate workspace representation; successful colour transfer

should not be conflated with colour being embedded in the MiniMesh tiles themselves.

For the independent-tile selections, the roof-reference and panorama results are important because they associate appearance observations with the same building identity used by the geometry pipeline. They provide registered inputs for ChordAtlas style generation. The separately retained appearance exports establish the availability of generated appearance artefacts, while reference registration, material completeness, and texture fidelity remain distinct evaluation questions.

5.4 Relationship among the evaluated systems

The project contributes at two interfaces to Sat3DGen. First, it extends the released large-image wrapper while retaining the pretrained DINOv3/tri-plane representation and one global density-domain surface extraction [2, 29]. Fractional placement and smooth overlap weighting address quantisation and hard contributor changes, while a second pass restores the colour capability available in single-image inference. The matched hard-box experiment isolates the effect of this fusion rule on boundary continuity and resource use.

Secondly, the mesh-space path introduces geographic placement, OSM/DSM constraints, building extraction, and ChordAtlas publication. This path remains relevant where independently extracted or cached meshes are the available interface. The two experiments therefore compare representation stages rather than treating mesh stitching as the only regional strategy. Independent reference geometry and street-level imagery are still required to evaluate metric and perceptual accuracy.

The retained London examples clarify that the systems address different subproblems. CityEngine generates controllable polygonal content from GIS shapes, attributes, assets, and CGA rules and supports scripted procedural runtimes [7, 16, 17]. Its retained export is a compact UV-mapped envelope whose openings are represented primarily in repeated facade atlases. Sat3DGen instead conditions dense geometry and vertex appearance on

overhead imagery. The retained ChordAtlas exports organise surfaces into UV material groups and appearance maps. The bridge connects these representations through stable identity, coordinates, and editable asset boundaries.

Table 5.1 summarises the distinct evidence and role of each system in this study.

Table 5.1: Evaluated systems, output representations, and their roles in the study.

System	Input and control	Output representation	Evidence and role
CityEngine / CGA	Platform support for GIS shapes, CGA rules, attributes, and assets; one initial shape in the retained case	Compact UV-mapped polygon export	Illustrative export for compactness and material organisation
Sat3DGen large-image	Rectangular mosaic, overlap schedule, and fusion weights	One density-fused surface with predicted vertex RGB	Directly matched OSM case; evaluated for continuity, colour support, and topology
Sat3DGen independent tiles	Nine image-conditioned meshes plus OSM/DSM operations	Stitched per-footprint assets with stored vertex RGB	Directly matched OSM case; evaluated for mesh-space assembly behaviour
ChordAtlas procedural appearance	Structured procedural surfaces and style controls	Semantic UV groups with diffuse, normal, and specular maps	Retained procedural appearance representation; not reconstruction truth or topology preservation
Hybrid bridge	Geographic selection, manifests, OSM/DSM context, and publication controls	Identified coloured OBJ, MiniMesh workspace, and registered appearance inputs	Integration contribution across learned and procedural representations

This is a partially matched representation case rather than a common reconstruction benchmark. The two Sat3DGen routes share bounds and footprint identities, whereas the CityEngine export lacks the common OSM identifier and ChordAtlas changes the asset representation. Topology and colour-variation statistics are therefore compared only between the two Sat3DGen paths. Across systems, the evidence supports conclusions about compactness, material organisation, available control mechanisms, and workflow structure.

5.5 Architectural implications of the manifest-driven design

The main value of the manifests is that they turn hidden assumptions into explicit system state. A request can record the intended geographic extent, required inputs, permitted tiles, completed stages, and reason for failure. This enables incomplete acquisition or processing to stop visibly instead of leaving an output directory that appears complete. Exact tile selection is especially important in a shared cache, where recursively consuming every available mesh can combine unrelated locations while still producing syntactically valid output.

However, a manifest is authoritative only when it binds outputs to their full provenance. Imagery, coordinate reference system, model checkpoint, inference options, processing configuration, code revision, and output hashes all influence the meaning of a cached result. The current design captures several of these fields but does not yet enforce all of them consistently. Stable geometric identity is therefore useful for request reuse, but it must be paired with configuration and source identity before cache reuse can imply reproducibility.

The cross-language boundary introduces a related transaction problem. Python can finish publication before Java has converted and loaded the corresponding asset. A stronger design would represent processing, conversion, and visible loading as separate acknowledged stages and would commit the final state only after the user confirms success.

5.6 Implications of the reliability findings

The large-image comparison shows that nominal overlap alone does not guarantee a continuous surface. With hard cropping and equal weights, the set of contributing windows

changes abruptly at regular intervals, turning local prediction disagreement into visible height steps. Fractional placement and raised-cosine weighting substantially reduce those steps without increasing the measured GPU peak, at the cost of modest additional run-time. Feathering smooths disagreement rather than resolving its cause; broad height bias shared by a window can therefore remain as a gradual surface variation.

The data audit shows that structural completeness and semantic validity are different properties. Files may be present while containing duplicates, cross-partition leakage, inconsistent coordinates, or candidates outside their intended spatial support. Dataset preparation should therefore validate identity, containment, and geographic separation in addition to file counts and schema shape. The London workflow is useful as prepared project data, but these findings prevent treating it as a validated training benchmark.

The mesh experiments reveal the same distinction between execution and correctness. A deterministic merge can still select non-contiguous inputs; a configured vertical threshold can remain ineffective if it is not applied by the matching predicate; and a DSM correction can produce a complete file while displacing surfaces that should not be treated as terrain. The relevant engineering principle is that every geometric assumption should have both an executable invariant and a regression case that can falsify it.

Building extraction also creates a trade-off between modularity and geometric quality. Footprint association produces independently addressable assets, which is valuable for editing and procedural processing, but cropping and capping do not by themselves guarantee closed, manifold, or semantically separated buildings. Topology and surface accuracy therefore require explicit post-processing and evaluation rather than being inferred from successful OBJ publication.

These observations lead to four general requirements for reliable integration:

- validate spatial and semantic relationships, not only the presence and shape of files;

- make tile membership and provenance authoritative at every stage;
- test every geometric assumption and specify how conflicting attributes are combined.

5.7 Contribution assessment

The contribution is a hybrid reconstruction and integration method rather than a replacement for either learned reconstruction or procedural modelling. It extends the released regional wrapper with continuous density and colour fusion, then connects the resulting surface to a shared spatial frame, building identity, external geographic context, modular publication, and executable data-transfer contracts. The London data analysis and controlled geometry tests expose assumptions that would otherwise remain hidden at these interfaces.

The project builds on Sat3DGen, ChordAtlas, ProceX, BigSUR, and FrankenGAN [2, 1, 12, 5, 6]. The project-specific contribution is the London data workflow, large-image density and colour method, geospatial mesh-processing method, ChordAtlas bridge, and evaluation developed around these foundations.

5.8 Threats to validity

5.8.1 Construct validity

Most reported measures concern structure and process: coordinates, manifests, vertices, faces, boundary discontinuity, topology, cache state, and publication state. These measures show whether transformations and contracts operate as specified, but they are not substitutes for geometric or perceptual accuracy. In particular, smoother seams, complete colour coverage, footprint overlap, and successful publication do not by themselves

establish façade fidelity or visual realism.

5.8.2 Internal validity and provenance

The large-image records bind the input plan, script hashes, processing parameters, and output hashes, but identify the model by repository name rather than one immutable revision shared explicitly by the density and colour stages. The current deployment also resolves those scripts through configured local paths. Packaging the scripts with the bridge and binding both stages to one model revision would make the experiment portable. At the consumer boundary, Python requires RGB for the large-image contract, whereas Java can still fall back from colour parsing; the same requirement should be enforced at both sides of the transaction.

5.8.3 External and statistical validity

The evaluation is concentrated in one London workflow. The fusion comparison uses a matched input, but city, morphology, acquisition, and checkpoint were not varied. External services also introduce network, credential, and licensing dependencies. Broader claims require a versioned multi-area evaluation with independent geometry and appearance references.

5.9 Priorities for further work

Three experiments follow directly from the findings. First, the matched fusion study should be repeated across multiple urban morphologies and model revisions, with independent surface and street-level colour references. A release-default hard-box run, the fractional-feather method, and relevant weighting ablations would separate boundary continuity from metric and perceptual accuracy.

Secondly, the global density path and the independent-mesh path should be run over the

same area and coordinate contract. Revised vertical stitching and DSM correction can then be assessed stage by stage for topology, height error, and surface preservation rather than compared through unrelated cached inputs.

Thirdly, the model revision, scripts, RGB requirement, MiniMesh conversion, and visible loading acknowledgement should become one portable publication contract. The resulting assets can then support a matched CityEngine/CGA authoring study that extends the present representation example with shared identity, inputs, and edit tasks, followed by a multi-user Ubiq study of interactive selection and editing.

6.1 Research outcome

This dissertation asked to what extent overlapping satellite-conditioned predictions can be combined into a coloured regional mesh and integrated with geospatial context in ChordAtlas. The study establishes both parts of this workflow. A rectangular large-image method fuses density predictions at sub-voxel positions, replaces abrupt overlap changes with smooth weights, and recovers colour on the fixed global surface. The resulting mesh is then placed in a shared spatial frame, associated with OSM footprints and DSM context, separated into modular assets, and published to ChordAtlas.

The matched London experiment showed substantially lower window-boundary discontinuity without increasing the measured GPU-memory peak, and the colour pass covered the complete surface without changing its geometry. The current on-demand integration retained that colour through coordinate conversion, DSM processing, footprint extraction, Java parsing, and building publication. Manifests, stable selection identities, cache checks, and explicit stage states make the process inspectable across the Python–Java boundary.

6.2 Principal contributions

The work makes three main contributions:

1. It extends large-image Sat3DGen inference to rectangular mosaics through sub-voxel density placement and smooth overlap fusion, and recovers global vertex colour in a memory-bounded second pass without altering the extracted geometry.
2. It defines a coordinate-explicit path from fused or independently assembled geom-

etry to footprint-associated ChordAtlas assets, preserving selection identity and vertex appearance while adding OSM and DSM context.

3. It introduces and evaluates manifest, cache, validation, and publication contracts across Python and Java, using data audits, matched fusion measurements, coordinate checks, topology inspection, and controlled regression cases.

These contributions provide a practical connection between learned urban reconstruction and procedural authoring. Sat3DGen supplies site-conditioned density and appearance, while OSM, DSM, and ChordAtlas add spatial identity, height context, modular asset boundaries, and an environment for further editing. CityEngine clarifies the complementary value of rule-driven semantics and controllability, while the retained ChordAtlas exports illustrate the representation difference between per-vertex colour and structured procedural material maps.

6.3 Implications and next steps

The evaluation identifies the next research priorities. Multi-area tests with independent geometry and appearance references are needed to determine whether smoother window transitions also improve reconstruction accuracy. The independent-mesh path requires corrected vertical stitching and more selective DSM processing, while the published building assets require topology repair. For deployment, density and colour should be bound to one immutable model revision, the large-image scripts should be packaged with the bridge, and readiness should include Java conversion and visible loading acknowledgement.

The current CityEngine export and retained ChordAtlas representation examples can be extended into a controlled authoring study with shared identity, inputs, and edit tasks, while Ubiq can test interactive and collaborative use of the resulting assets.

6.4 Closing statement

The dissertation connects bounded satellite predictions to a continuous coloured surface and an auditable urban-asset workflow. Its central value lies in combining learned density and appearance with explicit coordinates, geographic identity, and publication state, so that generated geometry can participate in a larger editable system. This provides a concrete foundation for evaluating regional reconstruction, procedural authoring, and collaborative XR with the same spatially identified assets.

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A | Implementation and Reproducibility Record

A.1 Evaluated components

The empirical study uses four principal implementations: the London data-preparation workflow, the Sat3DGen large-image extension, the active mesh assembly package, and the ChordAtlas integration layer. The project source code and submission package are available at <https://github.com/renjie22122-maker/FinalSubmit>. Runtime import and entry-point tracing identify the implementation used for each experiment. The published Sat3DGen, ChordAtlas, BigSUR, FrankenGAN, CityEngine, and Ubiq systems are referenced in the chapters where their methods or interfaces are discussed.

A.2 Component-test results

Table A.1 records the current component-level verification results. The Python suite exercises data prerequisites, coordinates, tile planning, geometry processing, manifests, cache behaviour, per-building publication, appearance references, and rollback. The Java validators exercise coordinate conversion, selection and cache contracts, OBJ slicing, vertex-colour transfer, MiniMesh publication, and appearance-bridge components.

Table A.1: Detailed component-verification results.

Test group		Result	Principal properties exercised
Python suite	integration	124/124 passed	Input validation, coordinates, geometry rules, manifests, cache state, publication, rollback, roof references, panorama acquisition, and large-image bridge contracts.
Large-image regressions	algorithm	9/9 passed	Fractional placement, coverage, feather continuity, colour-coordinate mapping, weighted RGB, and PLY round trip.
Java bridge validators		8/8 passed	MiniMesh conversion, selection identity, OBJ slicing, vertex colour, coordinate transfer, appearance integration, and GUI selection dispatch.
Controlled cases	geometry	2 counterexamples reproduced	Bottom-surface classification and the vertical constraint in vertex stitching.

The controlled geometry cases correspond to the predicate results discussed in Chapters 4 and 5. Quantitative figures were generated from the same evaluated artefacts and record their relevant populations in their captions or surrounding text.

A.3 Selected implementation excerpts

The following short excerpts connect Algorithms 1 and 2 to the submitted Python implementation. They are included verbatim from the project-authored scripts; the complete files, command-line interfaces, validation logic, and tests remain under `research/scripts`.

```
1 def raised_cosine_axis_weights(  
2     length: int,  
3     feather_width: int,  
4     taper_start: bool,  
5     taper_end: bool,  
6 ) -> np.ndarray:  
7     """Build a one-dimensional raised-cosine window.
```

```
8
9     Exterior mosaic boundaries deliberately remain at weight one.
Interior
10    tile boundaries taper from/to zero so contributors enter and leave
the
11    normalized blend continuously.
12    """
13    if length <= 0:
14        raise ValueError("weight-axis length must be positive")
15    if feather_width < 0:
16        raise ValueError("feather_width must be non-negative")
17    if feather_width * 2 > length:
18        raise ValueError("feather_width cannot exceed half the density
width")
19
20    weights = np.ones(length, dtype=np.float32)
21    if feather_width == 0:
22        return weights
23    if feather_width == 1:
24        ramp = np.zeros(1, dtype=np.float32)
25    else:
26        phase = np.linspace(0.0, 1.0, feather_width, dtype=np.float32)
27        ramp = 0.5 - 0.5 * np.cos(np.pi * phase)
28    if taper_start:
29        weights[:feather_width] *= ramp
30    if taper_end:
31        weights[-feather_width:] *= ramp[::-1]
32    return weights
```

Listing A.1: Raised-cosine axis weights used by the large-image density-fusion script.

```
1 def window_vertex_weights(
2     vertices: np.ndarray,
3     index: SpatialBins,
```



```
4     window: WindowSpec,
5 ) -> tuple[np.ndarray, np.ndarray]:
6     """Return vertex ids and exact positive 2-D contribution weights."""
7
8     length = int(window.row_weights.size)
9     if window.column_weights.size != length:
10         raise ValueError("Only square density windows are supported")
11     row_base = math.floor(window.density_row)
12     column_base = math.floor(window.density_column)
13     # Linear interpolation of the splatted edge can extend one global
14     cell
15     # beyond the discrete patch support. Exact weights remove false
16     positives.
17
18     candidates = query_spatial_bins(
19         index,
20         row_base - 1.0,
21         row_base + length + 1.0,
22         column_base - 1.0,
23         column_base + length + 1.0,
24     )
25
26     if candidates.size == 0:
27         return candidates, np.empty(0, dtype=np.float32)
28     candidate_vertices = vertices[candidates]
29     row_weight = splatted_axis_weight_at(
30         candidate_vertices[:, 0], window.density_row, window.
31         row_weights
32     )
33     column_weight = splatted_axis_weight_at(
34         candidate_vertices[:, 1],
35         window.density_column,
36         window.column_weights,
37     )
38     weights = row_weight * column_weight
```

```
33     selected = weights > 0.0
34     return candidates[selected], weights[selected]
```

Listing A.2: Exact support-weight selection used by the memory-bounded colour pass.

A.4 Model-example comparison record

The representation comparison is generated from the retained `modelsExample` directory. Only the `BigImageModel` and `satelliteImageModel` request/result packages provide directly matched identity: both name OSM way 369245408 and contain the same target bounds and footprint geometry. Their building OBJ files provide the geometry, topology, and vertex-colour evidence, while the result manifests provide the recorded DSM status and coverage reported in Section 4.8.

The CityEngine directory retains a CityEngine 2026 export log, OBJ, MTL, and three texture images, but no common OSM identifier. The two route-labelled ChordAtlas directories retain OBJ/MTL outputs and diffuse/normal/specular map sets; the OBJ files do not embed the footprint identity. The standalone ChordAtlas directory has no matching location or run manifest and is used only as an illustrative reference. The reproducible audit and software renderer are stored in `research/scripts`; their JSON/CSV outputs are stored in `research/results/models_example`. The rendered comparison figures use derived vertex colour or stored material maps and do not embed the source satellite mosaic.

A.5 Data and licence notes

OpenStreetMap-derived content requires Open Database Licence attribution [3]. The Environment Agency 2022 first-return DSM is used under the Open Government Licence and retains its stated coordinate and height reference [4, 40]. Google imagery is not

redistributed in the generated audit figures. Credentials found in local prototypes are omitted from the report.

B | Additional Matched London Building Comparison

B.1 Matched satellite-derived outputs

The `modelsExample2` record provides a second geographically matched case for the large-image and independent-tile routes. The two request packages contain identical selection bounds and footprint geometry for OSM ways 33719501 and 1455280501. The large-image route processes a contiguous 4-by-4 satellite mosaic through 100 overlapping inference windows, whereas the independent route assembles 20 separately generated meshes. Both result packages record successful DSM sampling for all vertices in the extracted building assets.

Table B.1 records the building-level geometry reported for both requested footprints. Vertical relief is the range within each saved model and is not a surveyed building-height measurement.

Table B.1: Recorded geometry metrics for the matched `modelsExample2` building assets.

Route	OSM way	Vertices	Faces	Relief (m)	Boundary / non-manifold edges
Large-image	33719501	122,119	242,601	37.75	1,617 / 0
Independent tiles	33719501	126,983	259,125	40.22	4,807 / 6,072
Large-image	1455280501	561	1,022	16.73	98 / 0
Independent tiles	1455280501	300	578	19.68	111 / 58

For the primary footprint, the large-image asset has fewer boundary and non-manifold edges than the independent-tile asset in this case. All four extracted surfaces remain open rather than watertight. The comparison therefore extends the case evidence in

Chapter 4 without treating mesh counts as a measure of reconstruction accuracy.

Figure B.1 compares the primary footprint under shared cameras and projection bounds. The renders use the RGB attributes stored by each satellite-derived route; no source satellite image is embedded in the figure. The generating script and machine-readable audit are retained under `research/scripts` and `research/results/models_example2`, respectively.

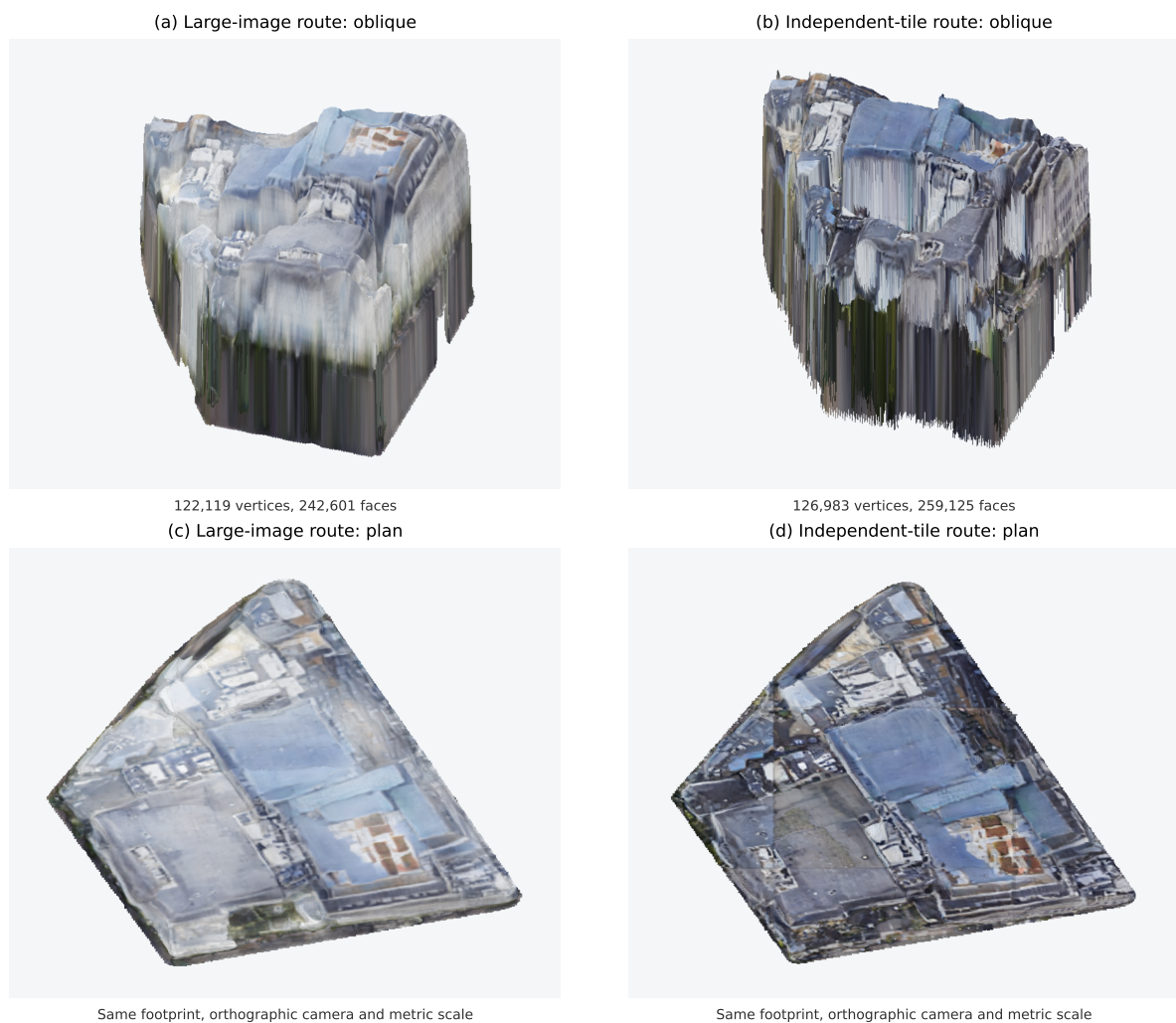


Figure B.1: Additional matched comparison for OSM way 33719501. Panels (a,c) show the large-image output and panels (b,d) the independent-tile output under shared oblique and plan-view bounds. Colours are the per-vertex RGB attributes stored by the respective satellite-derived routes.

B.2 Six representations of the selected building

The `modelsExample2` record contains six representations of the same selected London Trocadero building. They comprise the large-image and independent-tile Sat3DGen models, the corresponding ChordAtlas outputs, the CityEngine model, and the original ChordAtlas model. The two Sat3DGen result manifests identify the building as OSM way 33719501. The later OBJ/MTL exports belong to the same selected case, although they do not repeat the OSM identifier inside the model files.

The two Sat3DGen routes retain dense, irregular satellite-conditioned surfaces and encode appearance as per-vertex colour. Their corresponding ChordAtlas outputs replace vertex colour with UV-mapped diffuse, normal, and specular material structures while maintaining a similar horizontal building extent. The CityEngine model provides the most compact and regular envelope; its doors and windows are represented in shared image materials rather than as separate openings. The ChordAtlas packages contain independently materialised facade regions, including window-labelled materials and, in the independent-tile output, door-labelled materials. These differences describe geometry and appearance representation; greater face or material counts do not alone imply greater reconstruction accuracy.

Table [B.2](#) describes the stored geometry and appearance representation in every package. Face counts are reported as triangles after OBJ triangulation, except that the CityEngine row also retains its original polygon-record count.

Table B.2: Geometry and appearance records for the six representations in `modelsExample2`.

Representation	OBJ geometry	Appearance encoding	Method or stage
Large-image Sat3DGen	122,119 vertices; 242,601 triangles	RGB on every vertex; no UV material	Large-image route output
Large-image ChordAtlas	89,647 vertices; 728,314 triangles	69,041 UV records; 527 materials; 394 diffuse–normal–specular triplets	ChordAtlas output from the large-image route
CityEngine 2026	66 vertices; 23 polygon records (64 triangles after triangulation)	52 UV records; two materials and two diffuse images	CityEngine model
Independent-tile Sat3DGen	126,983 vertices; 259,125 triangles	RGB on every vertex; no UV material	Independent-tile route output
Independent-tile ChordAtlas	226,056 vertices; 2,597,150 triangles	156,686 UV records; 3,151 materials; 2,841 diffuse–normal–specular triplets	ChordAtlas output from the independent-tile route
Original ChordAtlas	149,021 vertices; 1,027,970 triangles	125,922 UV records; 572 materials; 424 diffuse–normal–specular triplets	Original ChordAtlas model

Figure B.2 presents the six outputs with a common oblique camera direction and metric display scale. Each model is translated to the centre of its panel without rescaling; the original ChordAtlas model is rotated by 180 degrees about its vertical axis so that the same side of the building is shown.



Figure B.2: Six representations of the same selected London Trocadero building. Panels (a,b) show the large-image Sat3DGen output before and after ChordAtlas processing; panels (d,e) show the corresponding independent-tile pair; panel (c) shows the CityEngine model; and panel (f) shows the original ChordAtlas model. The two Sat3DGen meshes display stored vertex RGB, whereas the CityEngine and ChordAtlas models display UV-mapped diffuse materials. All panels use a common camera direction and metric scale; models are centred without rescaling, and panel (f) uses the stated presentation rotation.

